

Received 12 March 2024, accepted 25 March 2024, date of publication 3 April 2024, date of current version 10 April 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3384272

SURVEY

A Comprehensive Survey on Revolutionizing Connectivity Through Artificial Intelligence-Enabled Digital Twin Network in 6G

MUHAMMAD SHERAZ¹, TEONG CHEE CHUAH¹,
YING LOONG LEE², (Senior Member, IEEE),
MUHAMMAD MAHTAB ALAM³, (Senior Member, IEEE),
ALA'A AL-HABASHNA^{4,5}, AND ZHU HAN^{6,7}, (Fellow, IEEE)

¹Faculty of Engineering, Multimedia University, Cyberjaya, Selangor 63100, Malaysia

²Faculty of Engineering and Science, Universiti Tunku Abdul Rahman, Kajang, Selangor 43000, Malaysia

³Thomas Johann Seebeck Department of Electronics, Tallinn University of Technology, 12616 Tallinn, Estonia

⁴School of Computing and Informatics, Al Hussein Technical University, Amman 11831, Jordan

⁵Department of systems and Computer Engineering, Carleton university, Ottawa, ON K1S 5B6, Canada

⁶Department of Electrical and Computer Engineering, University of Houston, Houston, TX 77004, USA

⁷Department of Computer Science, University of Houston, Houston, TX 77004, USA

Corresponding author: Teong Chee Chuah (tcchuah@mmu.edu.my)

This work was supported by Multimedia University Research Fellow under Grant MMUI/230013.

ABSTRACT The deployment of 5G has exposed capacity constraints in realizing the key vision of the Internet of Everything (IoE). Therefore, the researchers are exploring potentials of Digital Twin Network (DTN) in wireless networks. DTN is a novel technology to create virtual replicas of physical environment for testing, optimizing, and managing wireless networks. The integration of Artificial Intelligence (AI) and DTN appears to be a promising approach to address communication systems by providing an efficient environment for testing and improving AI models before deployment in real networks for effective network management, optimal resource allocation, and precise behavior prediction. Therefore, AI-enabled DTN in 6G represents a compelling avenue to address multifaceted challenges faced by wireless networks. In this comprehensive work, we offer a holistic survey that delves into the state-of-the-art approaches for AI-enabled DTNs in 6G. Firstly, we discuss the evolution of wireless networks and concept of AI-enabled DTN in 6G. Secondly, we discuss the role of AI-enabled DTN in 6G and driving advancements in fundamental components of 6G including resource allocation, caching, data offloading, and data security. Thirdly, we conduct a detailed discussion on key enabling technologies for realizing the capabilities of AI-enabled DTN in 6G. Fourthly, several applications of AI-enabled DTN in 6G are discussed for the practical relevance and significance in various industries such as smart cities, healthcare, and transportation etc. Finally, we provide lessons learned and highlight existing challenges and research directions to embark on further research efforts in the realm of AI-enabled DTN in 6G.

INDEX TERMS Digital twin networks (DTNs), 6G, artificial intelligence (AI), caching, resource allocation, data offloading, security, enabling technologies, unmanned aerial vehicle (UAV), mmWave, THz.

The associate editor coordinating the review of this manuscript and approving it for publication was Wojciech Sałabun¹.

I. INTRODUCTION

Wireless networks experienced remarkable technological advancement over several generations. Advanced Mobile Phone System (AMPS) in the United States of America

(USA) and Nordic Mobile Telephone in Europe were at the forefront of mobile voice-calling services when they were introduced in 1980. The inception of First-Generation (1G) wireless networks showcased the pioneering use of analog mobile telephony within the framework of cellular communication concepts. During the 1980s, these systems facilitated voice calling services for subscribers. Afterwards, a new mobile communication generation is launched approximately after every 10 years. Transitioning into the 1990s, the Second-Generation (2G) digital cellular networks replaced the analog 1G communication system. More than one billion people were served across the globe with an advent of the Global System for Mobile Communications (GSM) through digital voice telephony, Short Message Service (SMS), and low-rate data capabilities [1]. In 2000, amidst the explosive growth of mobile devices and the concurrent development of new mobile applications, there emerged a compelling need for significant enhancements in data speed. This led to the development of Third-Generation (3G) mobile communication systems that provided high speed data rate of Megabits/second based on Code-Division Multiple Access (CDMA) technology and Wideband Code Division Multiple Access (WCDMA) also known as Universal Mobile Telecommunications System (UMTS) [2].

Subsequently, the wireless network technology witnessed another significant leap with the commercial launch of Long Term Evolution (LTE) Fourth Generation (4G) mobile broadband services in Sweden and Norway in the 2009 [3]. The 4G was enabled by exploiting potentials of Multi-Input Multi-Output (MIMO), Software-Defined Radio (SDR), and Orthogonal Frequency-Division Multiplexing (OFDM). This breakthrough substantially enhanced data transmission speeds and network capacity, subsequently promoting the widespread adoption of mobile internet access and the proliferation of mobile devices [4]. This rapid evolution of mobile communication systems tend to occur approximately once every decade that is driven by the insatiable growth in users' data demands. The proliferation of mobile phones, Internet of Things (IoTs) devices, streaming services, and data-hungry applications has necessitated higher data rates and increased network capacity, thereby placing a significant burden on the existing 4G communication system. The Fifth Generation (5G) communication system builds upon the foundation established by its 4G predecessor, representing a substantial advancement in wireless communication capabilities, encompassing aspects such as cost efficiency, Quality of Service (QoS), and data rates [5]. 5G represents a transformative leap forward, bringing in not merely an incremental upgrade but a comprehensive re-imagining of mobile networks. 5G serves as the foundation for a diverse array of services and applications such as industry 4.0, immersive experiences of Virtual Reality (VR), the vast ecosystem of the IoTs, and the realization of automated driving [6]. Tables 1 and 2 offer valuable insights into the progression of wireless networks, spanning from 1G to 5G.

However, the research community acknowledges that the rapid proliferation of applications, including high-definition video streaming, Augmented Reality (AR), VR, and the IoT, has revealed the limitations of the existing 5G communication system in meeting the demands of these diverse and resource-intensive operations. Therefore, despite the implementation of 5G technology, researchers are envisioning a crucial necessity for technological advancement till 2030 to deal with the future massive data demands [7], [8], [9], [10].

A group of technologies for Network 2030 in International Telecommunication Union Telecommunication (ITU-T) standards sector is working on extension of network capabilities to cater future network services such as ubiquitous intelligence, holographic communications, and DTN [11]. The European Commission through its Horizon 2020 calls financing developments beyond 5G technology. Two projects ICT-52 Smart Connectivity Beyond 5G and ICT-20 5G Long Term Evolution are launched in 2020 for research projects on Sixth-Generation (6G) networks. These initiatives are advancing the development of wireless networks and emerging technologies to pave the way for highly data-intensive networks in the future. The European commission also invested in "Gigabit network connectivity" throughout the continent to accelerate digital technology throughput the Europe [12]. Moreover, Finland is also working on 6Genesis that is 6G-Enabled Wireless Smart Society and Ecosystem under the Academy of Finland's flagship initiative [13]. This program focuses on diverse key research areas including network intelligence, unlimited connectivity, and distributed computing. Currently, communication networks lack the real-time adaptability and predictive maintenance capabilities required to optimize the complex orchestration of 6G diverse components such as hardware infrastructure, protocols, and operating systems. The emergence of complex 6G networks presents a substantial challenge of optimizing these systems, which is hindered by the absence of real-time adaptive and predictive maintenance capabilities. This makes deployment, operation, and maintenance difficult for future wireless networks. Artificial Intelligence (AI) is a revolutionizing technique to provide predictions about wireless networks to the operators for handling anomalies and variations in the network. However, AI requires large amount of data for training and validations in wireless networks. The inherent requirement of availability and processing large amount of data, poses significant challenges in real-time applications due lack of computing resources and insufficient data for training, which can lead to inaccurate predictions.

The limitations arising due to the unavailability of training data and insufficient computing resources can be overcome through Digital Twin Network (DTN). DTN creates a virtual environment that mirrors the physical world. It ensures accurate virtualization of real-world physical objects and continuous synchronization with its physical counterpart to achieve optimal real time performance, proactive network maintenance, reduce downtime, efficient network planning,

TABLE 1. Evolution of wireless technology.

Generation	1G	2G	3G
Technology	Analog	Digital	Mobile Internet
Services	Voice Calls	Voice Calls, SMS, Basic Data Services	Voice Calls, SMS, Data Services, Mobile Internet
Data Speed	-	9.6 Kbps - 64 Kbps	200 Kbps - 2 Mbps
Bandwidth	30 kHz	200 kHz	5 MHz
Multiplexing Schemes	FDMA	FDMA and TDMA	CDMA
Core Network	Circuit-Switched	Circuit-Switched	Packet-Switched

TABLE 2. Evolution of wireless technology.

Generation	4G	5G
Technology	Mobile Internet	Mobile Internet
Services	Voice Calls, SMS, Data Services, Mobile Internet	Voice Calls, SMS, Data Services, Mobile Internet
Data Speed	100 Mbps to 1 Gbps	1 Gbps to 10 Gbps
Bandwidth	20 MHz	100 MHz
Multiplexing Schemes	OFDMA	OFDMA and SC-FDMA
Core Network	Packet-Switched	Packet-Switched

and data integrity [14], [15], [16], [17], [18]. An illustration of AI-enabled DTN in 6G is provided in Fig. 1. The versatility and potential impact of DT is realized in various domains including healthcare, renewable energy, and supply chain management etc. Therefore, DTs emerge as a highly promising tool to effectively realize both the technical and business aspirations of 6G networks [19], [20]. On the one hand, DT technology deployment requires a fast and reliable communication network for seamless data sharing between the physical and digital worlds. In this regard, 6G technology holds the potential to provide ultra-high data rates, ultra-low latency, and network reliability to achieve real-time data synchronization between physical systems and DTs [21], [22], [23]. On the other hand, an incorporation of DT into the 6G networks can offer significant advantages such as cost-effectiveness and enhanced performance by providing real-time insights and simulations to facilitate quicker decision-making and problem-solving. This can lead to faster response times and improved overall system performance. Consequently, researchers are engaged in the integration of AI and DTN for advancement of 6G networks. The AI-enabled DTN in 6G can simulate and optimize complex network configurations for rapid testing and fine-tuning of various components such as heterogeneous networks, edge computing platforms, advanced beamforming techniques, resource allocation, and dynamic spectrum sharing mechanisms. Therefore, it is necessary to present a systematic discussion of development in AI-enabled DTN in 6G to drive the research of both the academia and industry.

A. PREVIOUS WORKS AND MOTIVATION

The integration of AI and DTN in the context of 6G holds significant potential for optimizing communication systems. Therefore, it is essential and timely to render a survey article focusing on the deployment of AI-enabled DTN in wireless networks. However, existing survey articles do not focus on integrated AI and DTN deployment in wireless networks. Wu et al. [14] explored DT potentials

and provided a discussion on key enabling technologies and challenges. Moreover, authors also highlighted DTN applications for different purposes such as healthcare, aviation and urban intelligence. Barricelli et al. [20] presented a survey that investigates characteristics of DT technology and explored domains for DT applications. Moreover, the authors highlighted existing challenges and open research issues. Minerva et al. [24] presented a survey on DTN in IoT. The authors identified DT features pointing towards virtualization of physical objects.

Furthermore, Khan et al. [25] presented DTN key designing to enable 6G. Moreover, the authors discussed DT architecture trends including edge-cloud based twining, edge-based twining, and cloud-based twining. Shen et al. [26] delved in the 6G network architecture that is comprised of holistic network virtualization incorporating DT and network slicing and pervasive AI to foster service and user-centric networking. The authors provided detailed interplay between network slicing and DT model and data-driven approaches for management, and AI and virtualization to optimize network flexibility, adaptivity, scalability, and intelligence. Furthermore, Tang et al. [27] presented a comprehensive survey on DT in edge networks. The authors provided a detailed model of DT in edge networks, encompassing the design and deployment of the DT. Moreover, the authors highlighted key enabling technologies, applications such as IoTs and vehicular networks etc, and existing challenges of establishing DT in edge networks. Khan et al. [28] provided a survey on DT modelling, architecture, and frameworks in wireless networks. The authors presented DT design, devices, interface, and deployment level issues. The authors also highlighted data security and privacy issues in DT enabled wireless networks.

Alcaraz and Lopez [29] also surveyed data security threats in DT environment and performed classification of associated threats while taking into account operational demands and functionalities. In addition, the authors provided recommendations to achieve data security and discussed

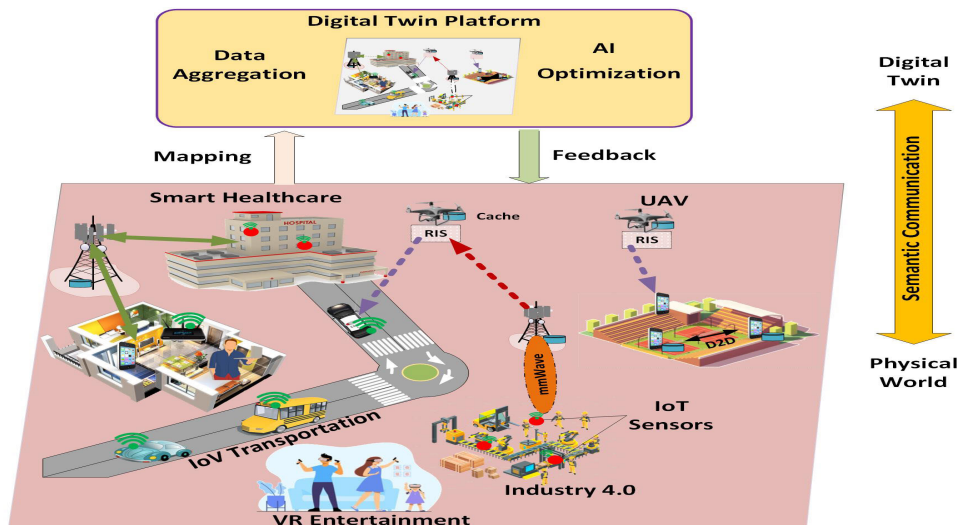


FIGURE 1. An illustration of AI-enabled DTN in 6G. The interplay of data caching and offloading is fortified by the security prowess of blockchain. DT is mapped through real-time information transfer from the physical environment. The intelligent decisions based on the real-time information, optimizes the functionality of the wireless network.

mechanisms to ensure data trustworthiness in DTs. However, DT technology is in its initial developmental stages, and its full realization necessitates the careful consideration of several critical aspects. These include cost, regulatory frameworks, interoperability, and standardization, all of which must be effectively addressed for its successful implementation. In our extensive literature review, we realized that there is a notable absence of a comprehensive survey that specifically addresses the integration of AI and DTN within 6G.

Hence, our aim is to present a comprehensive survey on the deployment of AI-enabled DTN for optimizing 6G. In Table 3, we provide a comparison of our survey with the existing notable surveys. This endeavor intends to offer our readers a broader perspective and a deeper comprehension of the utilization, advantages, and future research prospects within the domain of AI-enabled DTN in 6G networks. We seek to stimulate further research and innovation to drive the deployment of AI. We are expecting that our comprehensive survey will inspire our readers to generate novel research ideas and adaptation of AI-enabled DTN approaches for diverse applications in the realm of 6G.

B. SURVEY CONTRIBUTIONS

In our survey, we present a comprehensive survey that delves into the integration of AI and DTN in the context of 6G networks. This survey provides an in-depth understanding and exploration of the pivotal roles played by AI-enabled DTN in shaping the 6G landscape. We included notable research works focusing on fundamental communication techniques such as resource allocation, caching, data offloading, and security. Our survey encompasses key enabling technologies and a wide range of applications across various industries,

highlighting necessity to deploy AI-enabled DTN in the realm of 6G. We also highlight existing challenges and future research directions for advancement in the 6G networks. Our contributions can be summarized as follows:

- We conduct a systematic survey by embarking on a detailed discussion on the concept of AI-enabled DTN in 6G. We render valuable insights into the potential scope and deployment of AI-enabled DTN in 6G.
- We present an in-depth investigation of AI-enabled DTN employment in 6G to optimize fundamental components including resource allocation, caching, data offloading, and data security. The state-of-the-art developments are discussed aimed at harnessing the potential of AI-enabled DTN for 6G performance.
- We discuss key enabling technologies that support DTN in 6G framework for rendering understanding of technological infrastructure required for effective employment of AI-enabled DTN in the 6G.
- We discuss a wide range of applications that showcases how 6G can leverage AI-enabled DTN to revolutionize various industries supported by the 6G paradigm.
- We identify existing research challenges that need to be addressed and identify potential future research directions in the realm of AI-enabled DTN in 6G.

The rest of the paper is organized in the following manner. Section II delves into providing an insight of AI-enabled DTN and optimization of fundamental components of 6G. Section III investigated key enabling technologies to support DTN in 6G communication system. Section IV reviews applications of 6G DTN. Section V highlights existing challenges and potential future research directions. Finally, the concluding remarks are given in Section VI. The sections and subsections of this survey are depicted in Fig. 2.

TABLE 3. Comparison of our survey with existing surveys.

Ref.	AI	DTN	6G	Services	Key Enabling Technologies	Applications	Remarks
[14]	✓	✓✓	✓	✗	✓✓	✓✓	The authors presented a survey on development in DT technology and deployment in different applications.
[20]	✗	✓✓	✗	✗	✗	✓✓	The authors presented fundamental of DT technology, its implementations and applications.
[24]	✗	✓✓	✗	✗	✗	✗	This survey comprehensively discusses features and architectural models of DTN for IoT scenarios.
[25]	✗	✓✓	✗	✗	✗	✓✓	The authors presented a survey on DTN models and architecture trends for 6G.
[26]	✓✓	✓✓	✓	✗	✗	✗	This survey discusses holistic network virtualization and pervasive AI to meet service demands and user-centric networking.
[27]	✗	✓✓	✓✓	✗	✓✓	✓✓	A comprehensive survey of DT within edge networks, presenting a detailed model for designing and deploying DT. The survey emphasizes the importance of key enabling technologies and applications. Moreover, the survey highlights existing challenges associated with implementation of DTs in edge networks.
[28]	✓	✓✓	✓✓	✗	✗	✗	This survey focuses on DT modeling, architecture, and frameworks in wireless networks, addressing DT design, device issues, interface, and deployment level challenges.
[29]	✓✓	✓✓	✗	✗	✗	✗	A survey on data security threats in the context of DTs, offering threat classification, recommendations for data security, and mechanisms to ensure data trustworthiness.
Our Survey	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	An in-depth survey of AI and DTN implementation in 6G to optimize 6G services including resource allocation, caching, data offloading, security. Followed by detail discussion about key enabling technologies and application.

✓✓ In-depth, ✓ Superficial, ✗ Not addressed

II. AI-ENABLED DTN IN 6G

In this section, we offer a comprehensive overview of key elements and essential concepts essential for the successful integration of DT technology within 6G communication networks. To begin, we explore the concept of digital twinning, elucidating its significance and applications. Subsequently, we emphasize the profound implications of DTs on the evolution of wireless networks. Finally, we delve into the complexities of integrating AI and DTs, unveiling the synergistic potential of these technologies in the context of 6G.

In recent times, researchers have been actively exploring the deployment of Digital Twin Networks (DTNs) to optimize wireless networks. Recognizing the promising applications of DT technology in various domains such as healthcare, aviation, and manufacturing, the telecommunications industry is increasingly focusing on adopting DT technology [25]. 6G is anticipated to utilize advanced technologies to enhance network services capacity, reduce data access time, massive connectivity, and reliability [8]. However, 6G networks must overcome a number of challenges pertaining to spectrum allocation, privacy, and cost. DTN can play a crucial role in optimizing 6G ecosystem by accurate modeling and controlling network processes in real time.

Therefore, DTN deployment in wireless networks requires taking into account the complexity and scale of both the physical and logical assets of wireless networks such

as base stations, users, network services, and transport networks [30]. This move towards DT adoption is driven by the realization that advanced digital modeling and the twinning of physical objects and transport links within telecom systems offer real-time control over wireless networks. This not only enhances network performance but also ensures a superior Quality of Experience (QoE) for users [31]. Moreover, the integration of DT in wireless networks facilitates predictive network planning, efficient maintenance, and DT customization across network layers and processes. This holistic approach results in optimal network performance [32], offering the potential to significantly enhance operational efficiency while reducing costs, time, and resource requirements.

In wireless networks, the interactions between network infrastructure and end-users are dynamic and complex. Wireless networks face the challenges of diverse and evolving user behaviors, device preferences, and network conditions. This requires development of two DTs to provide a holistic view of the network in order to facilitate operators to optimize both technical aspects (infrastructure) and user-centric aspects (end-users' experiences) [30]. On the one hand, the first DT is dedicated to the network infrastructure, acts as a digital replica of the technical aspects of the wireless network. It meticulously models components such as base stations, antennas, spectrum allocation, and communication protocols. It provides network operators with a means to optimize these

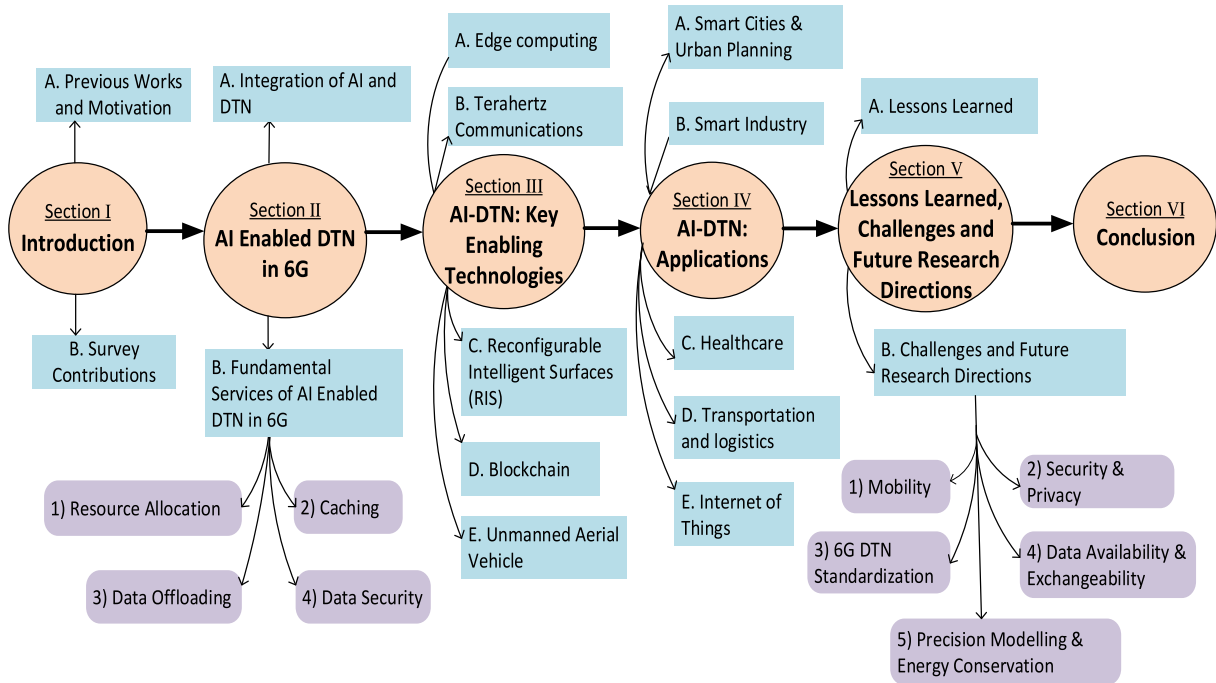


FIGURE 2. Survey structure of AI-enabled DTN in 6G.

technical facets for superior performance and efficiency. On the other hand, the second DT focuses on end-users and their experiences within the wireless network. This DT captures and analyzes user behaviors, preferences, and QoS expectations. Adopting a user-centric perspective, network operators possess the opportunity to refine their services with precision and aligning them to meet but users' expectations. This approach ensures that the network not only performs efficiently but also meets user expectations, which is essential for the success of wireless networks. Nonetheless, two DTs operate independently and data associated with one DT should not be exposed to the other DT. This isolation ensures data privacy and security, safeguarding sensitive information. The two DTs enable network operators to perform network testing in a virtual environment for optimal decision making in the physical 6G networks. This iterative and data-driven approach is pivotal in achieving optimal network performance and enhancing the overall wireless communication experience.

DT technology can enable twining of physical objects of wireless networks from Core Network (CN) to the end users to autonomously control and manage network elements [30]. There are numerous sub-twinning phases that may be used to achieve end to end twinning of the network infrastructure. For the 6G infrastructure layer, an E2E DT can be developed by first combining a number of DTs for the core network, transport network, and Radio Access Network (RAN) [33]. For instance, DT technology can optimize administration of 6G RAN through twining of base stations [33], where digital twining of base station should be comprised of base station

elements including antennas, cables, sensors, and power systems. The RAN management function collects a large amount of data regarding network performance, functionality, and operational characteristics. Then, AI can be employed to analyze all the data prior to feeding it to the DT of the RAN. This approach can enable network operators to optimize predictive maintenance, network operations and performance [34].

Furthermore, the DT can be devised to determine accurate status of the 6G network and alleviate network congestion by simulating in real time the network coverage area, predict QoS and QoE for the users [28]. For instance, DT can render visualization of the demography of coverage area, installed base stations, and end mobile users. This can enable network administrators to precisely model and predict the operational environment, wireless interfaces, data traffic, and assigned frequency. In the same vein, DT for Non-Public Network (NPN) such as vehicular networks can provide visualization of roads, road congestion, and traffic signals [35]. These applications leveraging DT attributes extend beyond network optimization and into user-centric domains. These attributes encompass personalized content delivery based on users' preferences, streamlining operational processes, and enhancing workflow efficiencies. DT technology holds the promise of not only optimizing data delivery but also improving the delivery of personalized content and value-added services, including data privacy and entertainment services, to end users. This approach can concurrently optimize the operational workflows and processes [28].

It is imperative to explore approaches to develop DTN for 6G networks. Firstly, it is necessary to accurately virtualize physical objects while capturing 6G network configuration, architecture, data traffic, and diversity [28]. Secondly, Standard Development Organization (SDO) defined models can be utilized to enable software and tools for DT to automatically develop and manage DTs with a minimum amount of time and resources [32]. Thirdly, it is necessary to merge DT with live DTs software models to ensure network operators can access real-time and accurate information to make optimal decisions for enhancing network intelligence and operations. Fourthly, it is imperative to deploy DT immediately and ensure quick execution. To attain execution rates faster than real-time, the DTs in a 6G network must employ cutting-edge simulation techniques such as quantum computing, parallel and distributed computing [36]. The DTs must possess the capability to seamlessly synchronize both live and simulated components in real-time. This synchronization is pivotal for establishing a dynamic and integrated test environment that incorporates DT equipment to improve environment monitoring, testing, and adaptability. Finally, DTs must include tools for correctly and effectively visualizing network traffic, underlying infrastructure, and end users to enable analysts and planners for detailed understanding of the 6G network functionalities [32]. These advantages allow network operators to spend less resources and time to maximize operational effectiveness including network planning, control and management, and network termination.

A. INTEGRATION OF AI AND DTN

The 6G communication system requires the collection of a massive amount of real-time data at DT to support ultra-low latency, high-speed data sharing, and high Quality of Service (QoS). The DTN can collect information from several domains encompassing radio access, core network, transport, and edge computing domains, layers, nodes (edge servers, base stations, and users), and communication links. 6G networks can utilize a range of data collection techniques including network traffic and QoS analysis to obtain data from various sources that exhibit diverse formats and structures. Therefore, it is imperative to execute precise data filtering, cleaning, correlation, augmentation, and labeling to maintain data integrity, privacy, and ensure compliance, which are pivotal for seamless 6G network operations. After data collection, all input data is normalized and transformed into a standard format for effective utilization by the DTN. This data uniformity enables the DTN to rapidly grasp the network performance metrics associated with each component of the 6G network.

In addition to efficient data utilization, an integration of DTN with AI approaches including supervised, unsupervised, semi-supervised, reinforcement learning, deep learning, and Federated Learning (FL) plays a pivotal role in efficient, reliable, and precise modeling of the physical

objects [37]. Then, all the data provided by the DTN can be utilized by the 6G network to evaluate diverse network deployment scenarios and operations. The data is leveraged for accurate prediction and testing of communication systems that can lead to development of mechanisms to achieve optimum network performance. All the decisions taken by DTNs are executed by 6G network after transforming data to a compatible format. Consequently, the 6G communication system can utilize recommendations rendered by AI-enabled DTN for optimal actions encompassing resource allocation, data privacy, and network operations.

Furthermore, the execution of optimization decisions in AI-enabled DTN 6G networks requires trusted and authorized Application Programming Interfaces (APIs) to facilitate efficient data sharing, interoperability, functionality, scalability, and data security [38]. The APIs are adaptive to dynamic network conditions and can adjust data flow and resource allocation in real-time. Hence, APIs serve as the bridge between the intelligent decision-making capabilities of AI and the real-time data and network control offered by DTNs to enhance the efficiency and effectiveness of the entire communication network. On the one hand, APIs within the DTN are designed to understand the specific data structures, protocols, and functions of the DT models. On the other hand, APIs in the 6G networks are tailored to interact with various network components, sensors, and devices. Then, API broker is utilized to ensure effective communication and interoperability between these two sets of APIs. The API brokers plays a crucial role in shaping a structured framework for interaction to enable smooth, automated exchanges of information, DTs, and various services such as network operation and maintenance between the DTN and the 6G networks. This approach can significantly reduce the complexity and costs associated with the interactions between AI-enabled DTN and 6G networks. Furthermore, it is essential to establish a standardized architectural framework to systematically create an intelligent and automated DT that is modeled for a 6G network. This framework defines how the DTN and AI-enabled 6G networks should interact and specify the processes, protocols, and interfaces. This standardization is critical for ensuring that the integration of AI and DTN in 6G networks is consistent, reliable, and capable of delivering the expected benefits. In the following subsection II-B, we explore fundamental and crucial elements including resource allocation, caching, data offloading, and security for developing an intelligent and adaptive AI-enabled DTN in 6G including resource allocation, caching, data offloading, and security.

B. ROLE OF AI-ENABLED DTN IN 6G FUNDAMENTAL COMPONENTS

In this section, we present state-of-the-art literature on AI-enabled 6G DTN and explore advanced techniques that harness AI's potential to enhance the functionality and efficiency of DTNs in 6G networks. These techniques

encompass resource allocation, caching, data offloading, and data security, which are fundamental in realizing the full potential of 6G DTNs across various application domains such as healthcare, smart cities, and manufacturing. We conduct an in-depth examination of these techniques and discuss how AI and DTN can pave path for unprecedented advancements in network performance, adaptability, and user experience within the evolving landscape of 6G connectivity. Subsequently, we discuss fundamental techniques of AI-enabled 6G DTN. Fig. 3 is an illustration of fundamental services provided by integration of AI and DTN in 6G. Moreover, a taxonomy of fundamental services of AI-enabled DTN in 6G is given in Tables 4 and 5.

1) RESOURCE ALLOCATION

The integration of AI-enabled 6G communication and DT can play a pivotal role in addressing the resource allocation complexities posed by the evolving wireless network architecture. As wireless networks transition to dense deployments, with the coexistence of macro and small cells sharing limited spectral resources, efficient resource allocation becomes a critical challenge [39], [40], [41]. This challenge is further exacerbated by 6G's landscape that is characterized by a diverse range of services with varying requirements. In the 6G complex landscape, it becomes imperative to maintain a balance in the distribution of resources. These resources are crucial for ensuring high performance and optimal resource utilization, especially in the context of data-intensive services like AR, VR, and 3D video streaming. However, the complexity arises from the diverse needs of computation-intensive applications, ultra-reliable low-latency services, massive IoTs connectivity, and other emerging technologies, each demanding tailored resource allocation approaches to meet their diverse operational requirements. The integration of AI-enabled 6G communication and DT can significantly drive progress in the 6G landscape by enabling intelligent synchronization between the physical and digital realms. This synchronization can be facilitated by AI-driven insights and real-time data from the DT to empower network operators to perform intelligent resource allocations decisions. An exploitation of AI-enabled DTN can enable dynamic and adaptive resource allocation based on evolving network conditions to support a wide range of services having stringent data and latency demands.

We discussed previously that the complex dynamics and diverse resources in AI-enabled 6G ecosystem can pose serious challenges of efficient resource allocation for computation-intensive and latency-sensitive DT services. These challenges stem from the evolving landscape of wireless communication, which is undergoing a transformative phase driven by recent advancements in sensing technologies and computing paradigms. For instance, with the advancement in the automotive sector, researchers are driven towards the development of intelligent transportation systems. Therefore, Chen et al. [42] developed a secure

autonomous driving system while ensuring data transmission secrecy and optimizing data transmission resources within smart cities concept. The authors devised a Convolutional Neural Network (CNN) and Decision Tree (DT) prediction models to seamlessly integrate network load balancing with a spatial-temporal graph convolution network for enhancing the connectivity and overall performance of autonomous cars. Furthermore, the authors proposed the Spatial-Temporal Graph Convolution Network (STGCN) algorithm for predicting complex traffic patterns in an urban environment. The obtained results demonstrate 92.7 % prediction accuracy, 80% data delivery rate, data leakage rates of approximately 10%. Thus, the proposed prediction model ensures minimal delay, network security, and efficient information sharing. For improving resource allocations, Huang et al. [43] introduced the DT Service Function Chain (DTSFC) approach and focused on the orchestration of dynamic DTSFCs to optimize resource allocation. The authors proposed a Collective Reinforcement Learning (CRL) approach inspired by collaborative human efforts to efficiently allocate resources for DTSFCs. CRL leverages cooperative knowledge exchange among agents highlighting its potential in reducing training duration, minimizing redundant model training, and enabling comprehensive models even in limited sample scenarios. The obtained results indicate that the proposed approach exhibits effectiveness in learning, displays adaptability to service delays, and offers a substantial throughput enhancement.

6G networks with high-speed, low-latency, and reliability are expected to provide an unprecedented connectivity, where Vehicular Edge Computing (VEC) can play a pivotal component in 6G [44], [45]. AI, edge computing, and DT can produce an intelligent and interactive mobility environment. Furthermore, the utilization of DT technology in Vehicular Edge Computing (VEC) enables vehicles to efficiently sense a wide array of information and subsequently transmit data to edge nodes through the robust communication framework known as Vehicle-to-Infrastructure (V2I) communications. This integration possess potential to empower VEC systems to harness the potential of digital twinning for improved data gathering and processing. Considering benefits of AI and DT in VEC, Liu et al. [46] exploited DT in Vehicular Edge Computing (DT-VEC) wherein diverse information is sensed by vehicles and transmitted to edge nodes through Vehicle-to-Infrastructure (V2I) communication channels. The authors proposed a cooperative sensing model utilizing a multi-class M/G/1 priority queue and V2I uploading model based on channel fading distribution and SNR threshold while taking into account consistency and timeliness of DTNs, data sensing, and communication resources. The problem is a bi-objective to minimize cost and improve system quality. The authors exploited Multi-Agent Multi-Objective (MAMO) deep reinforcement learning approach utilizing dueling critic network to assess agent actions based on the state and action. The obtained results demonstrate that the proposed approach provides significant reduction in network cost.

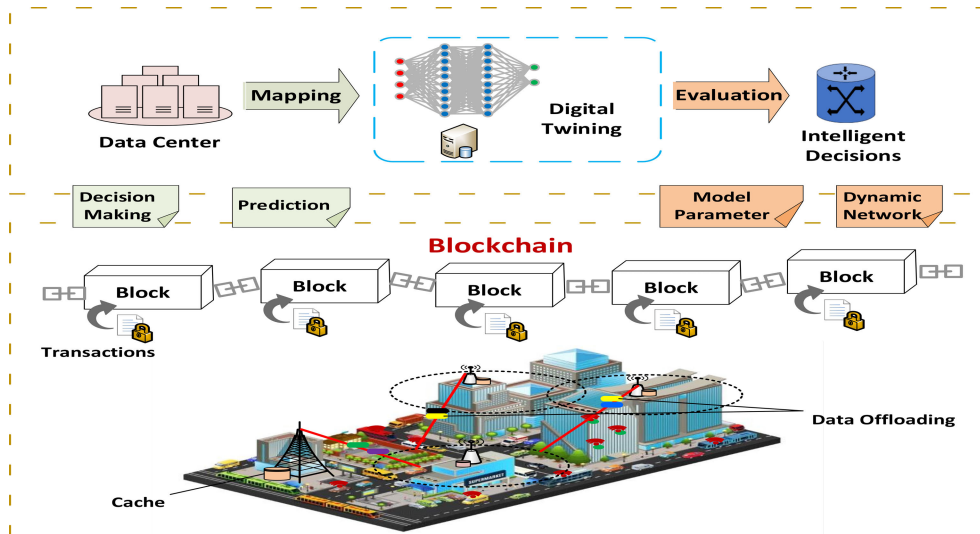


FIGURE 3. An illustration of fundamental services of AI-enabled DTN in 6G.

Moreover, in 6G communication system, IoT holds immense potential in order to enable highly intelligent services that can seamlessly adapt to complex real-time environments leveraging Network Function Virtualization (NFV) to dynamically provide Virtualized Network Functions (VNFs) for IoT devices. Within NFV-enabled IoT architecture, a service function chain (SFC) is composed of ordered virtual network functions (VNFs). Yet, the dynamic nature of IoT networks poses a significant challenge in terms of energy consumption during VNF migration and SFC reconfiguration. Therefore, Liu et al. [47] exploited potential of DT and proposed Bidirectional Gated Recurrent Unit (Bi-GRU) algorithm based on FL for accurate predictions. The proposed approach is efficient in determining the imbalance caused by reduced energy consumption and an impact of future evolving traffic dynamic traffic that lead to frequent VNF migrations. Moreover, the authors introduce a load balancing model to ensure network stability and minimize VNF migrations and energy consumption. A Distributed Proximal Policy Optimization (DPPO)-based Deep Reinforcement is used to optimize decisions in VNF migration based on prediction. The obtained results demonstrate that the proposed scheme is efficient in reducing VNF migrations and energy consumption.

2) CACHING

Caching has already shown tremendous benefits in wired networks to reduce latency and conserve energy. Therefore, it appears to be promising to implement the data caching in wireless networks in terms of optimizing network performance such as data access delay, QoS, and Cache hits. The key idea of data caching is placing contents close to the users by exploiting infrastructure caching (i.e., utilizing BSs) or infrastructureless caching (i.e., utilizing UEs) that can reduce delay and alleviate network overhead [48], [49]. However,

there are several concerns while designing efficient caching mechanisms including cache size constraint, dynamic data popularity profile, users' mobility patterns, and data access behavior. Therefore, caching requires processing of a large amount of information such as contents features, cache location, and cache techniques. An integration of AI and DTNs to optimize caching strategies appears to be a pivotal endeavor in shaping the efficiency and effectiveness of data delivery within the evolving 6G communication system. By applying AI techniques within the context of DTNs, the 6G networks can simulate and predict user behaviors, traffic patterns, and content popularity. This enables fine-tuning of caching schemes while considering limited cache resource, dynamic data popularity profile, and data access patterns.

Moreover, users are mobile in wireless networks and it is essential to accurately predict users' mobility trajectory to optimize data caching [50], [51]. The rapid expansion of Heterogeneous Vehicular Networks (HetVNs) offers potential for Internet of Vehicles (IoV) deployment and enhanced content distribution. However, designing effective caching mechanisms requires taking into account diverse vehicular user requirements, limited cache resources in Roadside Units (RUs), and dynamic connectivity patterns of Vehicular Users (VUs) and RUs. Hence, AI-enabled 6G DTN contains massive potential to optimize data caching by considering real-time dynamics and resource constraints of wireless networks. One effort of exploitation of AI-enabled DTN for data caching is performed by Hui et al. [52] that proposed an on-demand content delivery scheme for DT-enabled HetVNs (DT-HetVNs). The proposed scheme utilizes DT to simplify VU-RU interactions, where content requirements of DT of VUs (DT-VUs) are determined to enable collaborative content requests among VUs under the same RU. The interaction is modeled as a double auction game to determine content transaction prices. Then, a content

caching mechanism is formulated for each DT of RU (DT-RU) based on recommendations from different DT-RUs and content popularity. Afterwards, a collaborative caching algorithm is designed for optimal caching policy while taking into account cache capacity constraint. The obtained results demonstrate that the proposed scheme achieves low delay and high cache hits.

The advancement in 6G networks will facilitate intelligent vehicular networks that can promise ubiquitous vehicles connectivity within smart cities. This will redefine users' mobility while experiencing Vehicle-to-Everything (V2X) communication [53]. In this context, data plays a pivotal role in supporting a diverse range of applications aimed at ensuring road safety, elevating user experiences, and optimizing transportation efficiency within the evolving landscape of future smart cities. Data caching emerges as a crucial technique to optimize resource utilization by caching data to facilitate proximity mobile users in data access through the cache rather than putting burden over the core network. Hence, data caching can play an essential role in smart cities toward a futuristic intelligent and data-driven mobility. In addition, the rapid growth of smart vehicles and powerful applications has led to stringent content delivery requirements. Vehicular edge caching offers potential for timely content transmission, but challenges like limited cache capacity, dynamic network topology, and vehicle serving range can impact efficiency. To improve data access via limited cache resource, Zhang et al. [54] proposed a social-aware vehicular edge caching approach that exploits social awareness and DT technology. The proposed approach dynamically manages cache capabilities of RSUs and vehicles based on user preference similarity and service availability. Furthermore, the DT model is deployed to lead the concept of a vehicular cache cloud for enhancing content storage collaboration among cache-enabled vehicles in various traffic conditions. The system analyzes the content preferences of users, identifying similarities between their preferences, and strategically caches content accordingly. Then, the authors devised Deep Deterministic Policy Gradient (DDPG) learning to incorporate social model construction, cache cloud formation, and resource allocation. In this manner, the system refines its decisions and adapts varying user preferences and dynamic network conditions. The obtained results demonstrate that the proposed approach is efficient in achieving substantial optimal cache utility in vehicular networks.

3) DATA OFFLOADING

Data offloading is perceived as a potential approach for mitigating the data traffic burden from the backhaul [48], [62], [63]. As a consequence, improvements in network performance and data rates are realized, alongside reductions in latency. Data offloading enables to turn off or put some macrocell base stations into sleep mode by offloading some data traffic through other networks such as Wi-Fi,

Content Delivery Networks (CDNs), and Device-to-Device (D2D) communications. Hence, data offloading can mitigate the necessity for the continuous operations of all BSs. Consequently, the research community is exhibiting interest in the utilization of data offloading to reduce Operational Expenditure (OPEX) [64]. Additionally, researchers are also focusing on exploitation of data offloading to transfer computation-intensive tasks from mobile devices to computing resources like cloud servers or edge servers, which can enhance application usability and increased battery life [65]. However, the development of optimal offloading strategies is impeded by challenges stemming from factors such as mobility, dynamic wireless conditions, and the unpredictable nature of task arrivals.

In this context, an integration of DTs enables introduction of a transformative element in terms of leveraging decision making processes to predict and manage future tasks effectively. This proactive insight, empowers the strategic allocation of resources, seamlessly adapting to the evolving data demands and dynamics of the entire communication system. Considering the aforementioned problems and potentials of DT, Zheng et al. [55] proposed a DT enhanced framework that utilizes Deep Reinforcement Learning (DRL) approach and Asynchronous Advantage Actor-Critic (A3C) algorithm for optimal data offloading in the Internet of Vehicles (IoV). The proposed framework leverages a cloud-based DT as a software agent to access both global network insights via DT communication and historical vehicle data within the twin. Additionally, a learning scheme is integrated into the DT for predicting future demands and efficient computing resources allocations. The authors modeled the data offloading scheduling process as a Markov Decision Process (MDP) to minimize long-term costs encompassing task latency, energy consumption, and cloud renting expenses. It is evident from the obtained results that the proposed offloading scheme achieves quick convergence and significant reduction in cost.

To bring a paradigm shift in future mobile services, researchers are focusing on the potentials of integrating Mobile Edge Computing (MEC) and DTs. It is crucial to take into account the complex pivotal elements such as caching and interconnections between tasks that can degrade network performance. It is necessary for the research community to focus on optimizing resource allocation and computational offloading within the MEC-DT framework while considering constraints such as those pertaining to computation capacity, processing power, and cache space, particularly when handling resource-intensive services. One effort is conducted by Chen et al. [56] that focused on improving computation offloading and resource allocation for minimizing energy consumption. The authors proposed a DT-empowered MEC architecture that adeptly caters to the offloading of tasks by Mobile Users (MUs) while taking into account task inter-dependencies, service caching, and edge collaboration dynamics. The authors employed Asynchronous Advantage Actor-Critic (A3C) approach to optimize the allocation of

TABLE 4. Taxonomy of AI-enabled DTN to optimize fundamental components of 6G.

	Ref.	Objective	Proposed Approach	Outcome	Future Research Directions
Resource Allocation	[43]	Efficient resource allocation in AI-enabled 6G	CRL approach for DTSFC resource allocation	Effectiveness in learning, adaptability to service delays, increased throughput	FL should be exploited to enhance security in multiparty computation and homomorphic encryption can be leveraged to ensure secure collaboration while protecting sensitive data
	[42]	Secure data transmission and optimization	STGCN algorithm for predicting traffic patterns	92.7% prediction accuracy, 80% data delivery rate, data leakage rates	The development of a multi-channel media access control protocol is necessary to address growing demands of Internet of Vehicles (IoVs) services for robust connections.
	[46]	Minimize cost and improve system quality in VEC	MAMO deep reinforcement learning approach	Significant reduction in cost	Cooperation among edge nodes should be exploited and tested within real-world vehicular network environments.
	[47]	Minimize energy consumption during VNF migration	Bi-GRU algorithm for energy-efficient VNF migration	Efficient in reducing VNF migrations and energy consumption	Develop DT models for dynamic IoT conditions and resilient to network disruption while taking into account potential imbalance and evolving dynamic traffic
Caching	[52]	On-demand content delivery for DT-enabled HetVNs (DT-HetVNs)	DT-enabled on-demand content delivery scheme for cooperation between vehicles and road side units. Collaborative caching scheme algorithm for optimal caching	Low delay and high cache hits	Exploring joint caching mechanisms for users and relay users in HetVNs and harnessing blockchain technology to enhance data security and transaction privacy.
	[54]	Social-aware vehicular edge caching	DDPG algorithm for resource allocation	Optimal cache utility	Incentivizing smart vehicles to optimize vehicular edge data caching while considering complex traffic scenario with evolving vehicular topology.

computation offloading and resources in the DT-empowered MEC architecture. A3C consists of actor and critic. First, the actor performs decisions based on the current state of the system, then the critic evaluates the values of the decisions by estimating the expected future rewards. This combination of actor and critic provides optimal policies learning for improving energy efficiency. The obtained results demonstrate that the proposed approach is efficient in reducing energy consumption.

Moreover, the progress in urban informatics and vehicular intelligence has established connected smart vehicles as versatile edge computing platforms for diverse applications. However, the diverse array of smart vehicles with varying capacities and resource demands, coupled with unpredictable vehicular topology, possess challenges in realizing efficient edge computing. Zhang et al. [57] proposed AI-enabled DT of vehicular edge computing network to minimize task offloading cost within vehicular networks by dynamically adjusting

the DT state mapping mode. The proposed approach focused on optimizing edge service matching through cooperative assessment that devises multi-agent deep reinforcement learning approach for distributed computation task offloading and edge resource allocation. Furthermore, a coordination graph-driven vehicular task offloading scheme is proposed for efficient service matching and intelligent offloading scheduling across DT and physical networks. The authors utilized a real traffic dataset of taxi cabs in San Francisco Bay area to analyze the efficiency of proposed approach and the obtained results demonstrate substantial cost-effectiveness. In the realm of VEC, Dai et al. [58] also performed an integration of DT into VEC for prediction, estimation, and analyzing network states to minimize the task execution time by utilizing virtual models of vehicles and roadside units (RSUs). The authors proposed adaptive DT-based VEC network featuring AI-empowered closed loops for DT layer construction and VEC network management

TABLE 5. Taxonomy of AI-enabled DTN to optimize fundamental components of 6G.

	Ref.	Objective	Proposed Approach	Outcome	Future Research Directions
Data Offloading	[55]	DT-enhanced data offloading in IoV	DRL and A3C for neural network training	Quick convergence, cost reduction	Develop adaptive offloading mechanisms to dynamically deal with evolving network conditions such as data traffic loads and users' data demands
	[56]	Computation offloading and resource allocation in MEC	A3C-based optimization	Energy efficiency	Multi-agent RL and cooperative learning can be explored to optimize task offloading and resource allocation
	[57]	Task offloading in vehicular networks	Multi-agent DRL	Cost-effectiveness	Dynamic and context-aware service matching approaches are required for DTN leveraging real world information for efficient matching of edge services with massive vehicle pairs
	[58]	Adaptive DT-based VEC network	DRL-based adaptive offloading	Latency reduction	Development of DT encompassing not only vehicles and RSUs but also road infrastructure, traffic patterns, and environment conditions
Data Security	[59]	Enhance communication security and data privacy in DITENs	Blockchain empowered FL approach for security, Asynchronous aggregation scheme for efficiency of integrated scheme, and DT-empowered RL for scheduling relay users and spectrum allocation	Improved communication efficiency and data security	Privacy-preserving FL in DITEN should be explored while considering secure aggregation method, homomorphic encryption for data privacy while enabling collaborative edge computing
	[60]	Ensure data security and consistency in DT-assisted task offloading	DTTOS proposed for task offloading. To minimize power and latency, the problem is decomposed in two suboptimization models and DTA and DDQN are deployed, respectively	Data security, power conservation and low delay	Development of decentralized decision making algorithms enabling MECs to perform DT-assisted intelligent resource allocation, computation offloading, and data caching
	[61]	Improve FL in healthcare with reduced communication time	DTQFL algorithm is designed to train and update VQNN models	Enhanced VQNN accuracy, cost reduction	Investigating DT empowered advanced quantum algorithms and quantum circuit architectures for handling medical data processing and classification tasks

for ensuring network orchestration to optimize offloading. Then, a Deep Reinforcement Learning (DRL)-based intelligent algorithm is introduced to make adaptive offloading decisions. The obtained results demonstrate that proposed approach achieved a significant reduction in data offloading latency.

4) DATA SECURITY

Data security is of paramount importance in wireless networks due to pervasiveness and interconnections that raises serious challenges of users' information confidentiality and privacy. This requires advanced approaches to ensure information integrity, authenticity, and confidentiality.

AI plays a substantial role in wireless networks to safeguard data from unauthorized access, interception, tampering, and other malicious activities to enhance data security. However, AI algorithms require extensive and diverse datasets to achieve optimal learning and determine potential threats, where it is hard to obtain datasets to ensure privacy. Furthermore, AI models need continuous updates to stay effective against evolving security threats. Deploying these updates directly in a live network environment can be risky, as they may disrupt operations or inadvertently introduce vulnerabilities. In this case, DTN appears to be a potential approach that can create a virtual environment and enable simulation of security scenarios. DTN can offer a safe testing ground to assess the impact of AI model updates before implementing them in the real network to avoid potential risks. Thus, considering the immense potential of AI-enabled DTN, it is evident that an integration of AI and DTN can enable learning from a broader range of potential threats while maintaining data integrity.

DTs can be utilized to maintain a seamless and secure wireless network. Therefore, DTs can be developed by devising cloud-based computing architecture by collecting extensive data and execution of AI on a centralized server. However, the centralized processing approach incurs massive data processing burden that can lead to data privacy issues. Hence, a novel approach is required to apply AI in a distributed manner while maintaining data security and users' privacy. Addressing the crucial matter of data security in offloading processes, Lu et al. [59] proposed a blockchain-empowered FL scheme to improve communication security and data privacy within DT Edge Network (DITEN). Additionally, the authors introduced an asynchronous aggregation mechanism and DT-empowered reinforcement learning to optimize user scheduling and spectrum resource allocation. The obtained results demonstrate substantial gain in communication efficiency and data security.

It is necessary to take robust measures such as encryption protocols, access controls, and blockchain technology to ensure data integrity, avoid unauthorized data access, and cyber threats. These advancements are pivotal in enhancing the overall security of offloading to achieve reliability and confidentiality of data transfers within the network. In this vein, Liu et al. [60] proposed DT-assisted task offloading scheme (DTTOS) that involves Mobile Edge Server (MES) selection and intelligent task offloading. The authors employed blockchain and Channel State Information (CSI) for selection of reliable to ensure data security and consistency. The problem complexity is managed through a two-step optimization approach using Decision Tree Algorithm (DTA) and Double Deep-Q-learning Network (DDQN). The obtained results demonstrate that the proposed approach achieve data security, power conservation, and low delay.

In the realm of healthcare, numerous innovative solutions are emerging, particularly with the integration of AI. While this integration holds the potential for transformative

outcomes, it is important to note that AI approaches possess inherent challenges such as data privacy concerns, extended training periods, and delayed update processes. One AI-DT approach is developed by Qu et al. [61] for healthcare applications. The authors proposed a DT-assisted Quantum FL (DTQFL) algorithm that leverages mobile network to develop a real-time DT of patients using Internet of Medical Things (IoMT) devices. This significantly reduces communication time in FL while training Variational Quantum Neural Networks (VQNNs). DTQFL's approach involves synchronized training and updating of VQNNs for patients with specific diseases, achieving accuracy comparable to centralized algorithms. The obtained results demonstrate that the personalized training enhances the VQNN accuracy and cost is minimized.

III. AI-DTN: KEY ENABLING TECHNOLOGIES

In this section, we embark on a comprehensive exploration of cutting-edge key enabling technologies that drive a transformative paradigm shift in the realm of AI-enabled 6G DTN. We endeavor to render an in-depth discussion on pivotal evolutionary key enabling technologies including Edge Computing, Terahertz Communications, Reconfigurable Intelligent Surfaces, Blockchain, and Unmanned Aerial Vehicles (UAVs). We will discuss how these key enabling technologies leverage the potentials of AI-enabled DTN to converge and shape the 6G communication system. Table 6 summarizes the role of key enabling technologies towards AI-enabled DTN in 6G.

A. EDGE COMPUTING

Edge computing is a technology that efficiently manages data processing, storage, and network resources located between data sources and data centers. However, despite the numerous benefits of DTN in 6G, edge computing presents significant challenges, including users' data privacy, energy conservation, and latency. Edge computing appears to be a promising technology capable of providing data processing in close proximity to end-users, effectively mitigating latency and data privacy concerns. Furthermore, it facilitates task offloading to end-users, thereby contributing to the minimization of energy consumption. Consequently, due to its inherent scalability, support for real-time decision-making, and network resilience, edge computing aligns seamlessly with the requirements of DTNs. However, it is crucial to recognize that the limited computing resources of edge devices can potentially result in latency issues, especially when dealing with large task queues. To address this challenge, a collaborative edge computing approach can be employed among network edge nodes to enhance Quality of Service. Collaborative edge computing efficiently allocates computational tasks among available edge nodes, ensuring a balanced workload and enabling low-latency computing services in time-sensitive applications, such as intelligent transportation and medical services. Furthermore, DTN tasks are computationally intensive and demand high

computational capabilities [66]. In this context, Cooperative end-edge-cloud computing can effectively manage massive computational tasks and real-time modeling of DTN. This innovative approach involves real-time data processing at edge servers, with cloud servers providing substantial computational power and facilitating seamless information integration. This approach fosters a cohesive and efficient synergy between edge and cloud resources, effectively addressing the computational demands of DTNs while ensuring real-time responsiveness and data integration.

Moreover, it is essential to delve into the management of data heterogeneity within the cloud through real-time interaction between edge users and cloud. This requires understanding the complexity of the dynamic process within DTN. When edge users encounter limitations in data storage capacity, then cloud can participate in data storage and transfer data to the client via wireless network. This approach conserves edge nodes energy and alleviates data storage constraint. Hence, DTN can benefit from collaborative end-edge clouds to achieve low latency, high computations, and continuous environment update. To bridge the gap between the intricacies of managing data heterogeneity within the cloud and the convergence of physical edge networks with DTNs, Lu et al. [59] proposed an asynchronous aggregation scheme and exploited Reinforcement Learning (RL) to schedule data relay users and allocating bandwidth resources for optimizing system efficiency in DITEN. Moreover, the authors exploited a blockchain empowered FL approach to achieve data security and privacy in DITEN. The obtained results demonstrate that proposed mechanism achieves data communication efficiency and security. Then, Lu et al. [67] exploits communication-efficient FL scheme for DITEN in industrial IoT to reduce communication overhead and transmission energy cost while ensuring data privacy. The authors utilized FL for developing DT model of IoT devices based on information from the physical environment. Then, asynchronous model update approach is devised to alleviate communication overhead burden and reduce communication cost. Furthermore, the authors leveraged Deep Neural Network (DNN) model for optimal allocation of communication resources that provides improved communication efficiency and reduced energy expenditure.

Sun et al. [68] focused on minimizing data offloading latency while taking into account migration cost of accumulated service and mobility. To tackle this problem, the authors also used DITEN that employs DTs of edge servers for estimating states of edge servers and DT of Mobile Edge Computing (MEC) system helps to obtain training data to optimize data offloading decisions. A Lyapunov optimization approach is exploited to deal with the long-term migration service cost constraint and converts problem into a multi-objective dynamic problem, which is solved by ML approach of actor-critic deep reinforcement learning. The obtained results demonstrate a significant reduction in data offloading latency and service migration rate, and achieved substantial data offloading success in 6G system. As we

know that IoTs is crucial is in 6G that hinders achieving high network efficiency due to evolving network dynamics, resources constraints and heterogeneity. Dai et al. [69] proposed a concept of DTN to model IoTs network topology and stochastic task arrivals to handle edge computation and resource allocation in IoTs. The authors devised a Lyapunov optimization technique to convert a stochastic problem to a deterministic per-time slot problem. Then, an asynchronous actor-critic approach is deployed to determine the best stochastic computation offloading and resource allocation strategy. The obtained results demonstrate a significant in system cost.

B. TERAHERTZ COMMUNICATIONS

It is envisioned that the 6G communication system will be consisting of diverse devices. This massive interconnected communication system requires seamless and accurate sharing of large amounts of information to ensure that the DT is an exact copy of the real world. Here, THz appears as a pivotal enabler offering high-speed data transmission to handle substantial data loads. This immense data capacity of THz can handle the massive data traffic of 6G DTN to accommodate real-time communication and complex data processing. Moreover, THz vast frequency range can enable rendering services to 6G ecosystem including mobile devices, machines, IoTs, virtual and AR devices, and smart home devices etc. There are a number of issues that need to be resolved in the design and implementation of THz communications. THz connections are susceptible to deep fading and signal blockage that lead to unreliability of THz links and degradation in QoS. Moreover, due to narrow beamwidth THz waves have restricted coverage and multi-user support. Researchers are actively exploring ways to leverage the unique properties of THz waves to overcome these challenges and unlock the potential of THz communication technology for future applications, including in the context of 6G networks. Pengnoot et al. [70] proposed THz signals guiding mechanism, where DT is utilized for modelling, predicting, and controlling signal transmission path. Therefore, tunable metamaterial walls are deployed between serving BS and user for optimal signal propagation path selection. The authors deployed advanced image processing systems to provide contextual updates to the DT. The obtained results demonstrate that the proposed approach maintains connectivity while using a minimum number of reflector strips and antennas.

Moreover, VR is rapidly gaining prominence but demands ultra-high data rates and ultra-low latency. The high bandwidth of THz technology can enable streaming high quality VR content without latency. Researchers are developing mechanisms to exploit THz to enable VR in 6G. However, it is necessary to take into account that THz communication has a high tendency to be blocked due to its short propagation range and the narrow beamforming [71], [72], [73]. Consequently, the deployment of THz for VR presents

two essential challenges. The first one involves indoor mobile VR users dynamically obstructing THz links with their moving bodies, leading to significant disruptions in indoor VR experiences [71]. The second challenge is to achieve seamless interactions between indoor VR users and the actual real world, requiring precise user positioning to generate corresponding VR images [74]. Tao et al. [75] focused on minimizing the delay to serve users in the VR system. The authors proposed a DT communication networks incorporating THz and visible light communication (VLC). VR images are transmitted by the Macro Base Station (MBS) to the Small Base Station (SBS). Then, the SBS relays images to users through THz links. The VR users' positions are precisely modeled by VLC positioning system that enables the BS to generate VR images. This problem is transformed into a linear problem and the Big-M approach is utilized to solve delay minimization problem. The obtained results demonstrate a reduction of 33.2% in data transmission delay in the VR system. The deployment of DTNs can experience challenges due to massive twin data and constrained network resources. Zhang et al. [76] introduced a novel DT architecture for THz wireless networks to enhance 6G networks, which includes user, service, and DT layers. The authors proposed distributed message propagation algorithm based on Graph Neural Network (GNN) to produce user association and power assignment scheme of DTN. The proposed scheme utilizes node initial characteristics in graph for aggregation of adjacent nodes information to determine nodes hidden relations. The obtained results demonstrate a significant improvement in DT network mean rate for THz band.

C. RECONFIGURABLE INTELLIGENT SURFACES (RIS)

RIS technology is expected to improve the wireless communications. RIS are composed of numerous passive reflecting components that can intelligently manipulate signals propagation and distribution through precise adjustments of phase shifts. RIS is emerging as a key enabler to substantially improve data rate, network coverage, and mitigate interference [77]. Therefore, RIS seems to be promising in 6G-DTN for offering a variety of intriguing applications such as indoor connectivity, THz communications, and healthcare etc. For instance, in a multi-cell 6G DTN scenario, RIS possess the capacity to simultaneously enhance the received signals from serving BSs that may reduce intercell interference. Moreover, RIS is efficient in increasing data offloading rate. The data offloading volume from devices to edge servers is dependent on the wireless link gain. Through virtual array gain and reflective beamforming, RIS can offload more data to edge servers to process data in less time. Therefore, researchers are conducting efforts to deploy RIS in the 6G communication system. Sheen et al. [78] proposed the Environment-Twin (Env-Twin) model for RIS supported wireless networks. The authors utilized the proposed model for RIS reflection beamforming from position attributes of a receiver without

beam training or channel estimation. The proposed model learns the mapping function between RIS configurations and achievable rates for receiver locations without explicit channel estimation or beam training. After mapping, the proposed model can predict RIS configurations for each new receiver locations and deep learning is devised. The obtained results demonstrate that the proposed model can efficiently provide RIS configurations with small overhead compared to the approaches using channel estimation information at the RIS active elements.

Furthermore, Internet of Vehicles (IoVs) emerges as a transformative mechanism to revolutionize intelligent mobility. IoV will be an integral part of IoT ecosystem that envisions seamless integration between smart devices, vehicles, and transportation systems. However, traditional wireless systems have limited resources and it becomes difficult to maintain high quality connections in densely populated and dynamic environments. Therefore, there is a serious challenge of ensuring efficient and reliable wireless channels between mobile vehicles and infrastructure to achieve low latency, real time decision making, and maintain high transportation efficiency. RIS appears to be a promising solution to address these challenges. RIS are intelligent reflecting surfaces that can alter the wireless propagation environment by adjusting the phase shifts of incident signals. Yuan et al. proposed the DT driven Vehicular Task Offloading and IRS Configuration framework (DTVIF) for effective learning, monitoring, and management of Internet of Vehicles (IoVs). The authors adopted edge computing and IRS for augment computation and enhancing data transmission between vehicles and edge servers. DT is deployed for real-time data aggregation and virtualization of physical environment for optimal decisions. In this work, Two-Stage Optimization for Jointly Optimizing Task offloading and IRS Configuration (TSJTI) algorithm based on DRL and Transfer Learning is proposed to manage energy consumption and delay. First, Double Deep Q-Learning Networks (DDQN) is used for optimizing offloading decisions. Second, optimal IRS configuration is determined based on Deep Deterministic Policy Gradient (DDPG). The obtained results demonstrate a substantial improvement in task offloading delay minimization and energy consumption.

D. BLOCKCHAIN

The advancement of edge intelligence heavily relies on the utilization of edge devices for data processing, analysis, and operating applications that play crucial roles in executing learning tasks such as classification and regression. However, this dependence on edge devices exposes edge intelligence to vulnerabilities because malicious entities can attack to steal data and manipulate parameters during data transmission and training processes. Furthermore, a significant security concern emerges from the presence of untrusted edge devices within collaborative training scenarios, which can exploit sensitive information exchanged during gradient exchange. This elevates the likelihood of severe data breaches.

TABLE 6. Role of key enabling technologies towards AI-enabled DTN in 6G.

	Ref.	Objective	Proposed Scheme	Utility
Edge Computing	[59]	Improve security and efficiency of edge computing	Blockchain empowered FL scheme	Achieved high efficiency and accuracy
	[67]	Real-time data analysis and communication cost minimization	FL asynchronous model update scheme	High communication efficiency and reduced energy cost
	[68]	To reduce offloading latency	Lyapunov dynamic queue optimization method and actor-critic deep reinforcement learning	Effective reduction in latency and task failure rate while maintaining low cost
	[69]	Stochastic computation offloading and resource allocation	Lyapunov optimization technique and asynchronous actor-critic algorithm	System cost is significantly reduced
THz	[70]	Design, predict, handle signal propagation characteristics	THz Potential Field (THzpf) inspired by Artificial Potential Fields	Maximized SNR and maintained connectivity with minimum use of antenna and reflector strips
	[76]	DT architecture for THz wireless networks	Distributed message propagation algorithm based on Graph Neural Network (GNN)	Minimized mean rate for THz band
RIS	[78]	Capturing characteristics and interactions	Env-Twin framework and DL approach	Achieved near-optimum data rate
	[79]	Focused task offloading and IRS configuration	TSJTI algorithm, DQN, and DDGP	Minimized energy consumption and delay
Blockchain	[80]	To guarantee secure and reliable traceability	Smart contracts and decentralized storage	High data security is achieved
	[81]	FL and blockchain technology in wireless networks	FL and multi-agent RL	Reduced latency and learning convergence
	[82]	Blockchain empowered adaptive asynchronous FL (FedTwin)	Adaptive asynchronous FL (FedTwin)	Achieved privacy and substantial accuracy
UAV	[83]	Optimizing task offloading and energy consumption minimization	DDQN algorithm	Substantial reduction in energy consumption
	[84]	DT enabling BS and UAV assisted Mobile Edge Computing	MAPPO algorithm based on DRL	Substantial reduction in energy consumption
	[85]	Improving task completion ratio in multi-UAV systems	DQN	Improved task completion and conserved energy
	[86]	A smart navigation framework to minimize flight time	DRL	Extended flight duration and conserved energy

To establish a secure and dependable edge intelligence ecosystem in 6G, it is imperative to propose and implement effective security solutions. The 6G communication system involves the convergence of distributed communication, computing, storage, sensing, and controlling resources to deliver various services such as pervasive computing, AI, blockchain, and DT integration. Integration of 6G DTN and blockchain appears to be promising as a strong driving force that can ensure security guarantees, trusted traceability, accessibility, and immutability of transactions in the 6G DTN environment. For instance, the decentralized data learning is a innovative approach in edge intelligence systems. The decentralized data learning emerges as an innovative approach in edge intelligent system wherein blockchain ledgers play a pivotal role in validating the computational processes of edge intelligence models. This entails empowering edge blockchain peers to collectively and

autonomously verify accuracy of a shared AI model while maintaining the confidentiality of the underlying datasets.

Blockchain is fundamentally a chain structure of data blocks arranged chronologically, serving as a tamper-proof distributed database that ensures the security of each link through cryptography in a decentralized manner. It comprises peer-to-peer networks, distributed storage, consensus mechanisms, cryptography, and smart contracts, offering various advantages such as decentralization, tamper resistance, anonymity, public verifiability, and traceability. Nevertheless, for seamless integration of blockchain into DTN, it is essential to consider that computation intensive processes can introduce latency in blockchain technology. This raises a serious concern that how blockchain's inherent security features can be exploited without compromising the DTN critical features such as low latency and computational efficiency.

Considering the nature of blockchain technology and its potential implications for DTNs, it is essential to delve deeper into its integration challenges. Currently, DTN technologies tend to be centralized that lead to deficiencies in ensuring data provenance, auditability, and traceability. Furthermore, it introduces vulnerabilities in terms of data security and the potential for unauthorized tampering, particularly concerning transactions, logs, and historical records. Hence, the integration of blockchain into DTNs necessitates a thorough evaluation to ensure that its inherent security features align seamlessly with the critical requirements of DTNs, notably low latency and computational efficiency. To deal with this issue, Hassan et al. [80] proposed a blockchain-based approach for creating DTs that guarantees secure and reliable traceability, accessibility, and immutability of transactions, logs, and data provenance. The proposed approach utilized smart contracts to govern and monitor transactions that are initiated by the involved participants during the DT creation process. A decentralized storage is employed using interplanetary file systems to optimize data security and accessibility. The authors conducted an analysis of the proposed approach using the SmartCheck tool, and the outcomes clearly indicate a high level of security satisfaction. Additionally, this innovative approach introduces enhancements in process management, preserves comprehensive records of registered owners, and provides a detailed account of process history. Lu et al. [81] exploited FL and blockchain technology in wireless networks. The authors proposed DT Wireless Network (DTWN), where a blockchain-enabled FL mechanism is employed to improve data reliability, security and privacy. Then, multiagent reinforcement learning technique is used to achieve learning accuracy. The obtained results demonstrate a significant reduction in latency and desired learning convergence is achieved. Qu et al. [87] proposed block chain empowered an adaptive asynchronous FL (FedTwin) approach to preserve privacy and decentralization of DTNs. The authors designed Proof-of-Federalism (PoF) algorithm for autonomous DTN. Moreover, generative adversarial network enhanced differential privacy is devised during local training of each DT for privacy protection of model parameters. In addition, isolation forest is employed for filtering falsified DTs. Then, Markov Decision Process (MDP) is used in global collection phase for DT selection for achieving adaptive asynchronous aggregation and rollback approach is used to avoid global falsified models.

The convergence of DT and 6G brings forth a range of advantages such as enabling advanced services, real-time interactions, and seamless integration of IoT devices. This integration contributes significantly to the widespread adoption and success of IoT-enabled appliances in the 6G era. Despite these promising prospects, IoT-enabled home appliances are still in their early stages of development, particularly from the perspective of customers, which poses a challenge to their widespread market penetration. To address customer reservations and facilitate greater acceptance of

IoT appliances, it becomes essential to design business models that prioritize the needs and preferences of users over corporate interests. Altun et al. [82] proposed a novel model that specifically focuses on human-centric business models instead of corporate-centric ones. In this context, the model ensures that the absolute ownership of fog-located DTs of home appliances is granted to their respective owners, fostering a sense of control and empowerment. Then, the proposed model further enhances human-centric services and applications by incorporating blockchains and blockchain-enabled clouds that allows customers to have complete ownership rights over these fog-located DTs facilitates the liberalization of both the DTs themselves and the appliances associated with them. This approach ultimately uncovers the true value of human-centricity and fosters collaborative interactions between various interconnected things-to-things.

E. UNMANNED AERIAL VEHICLE (UAV)

UAVs are envisioned to provide substantial extension in network coverage while serving as aerial BSs. The deployment of UAVs as BSs is a cost-efficient approach to provide wireless communication to geographically challenging areas containing limited infrastructure such as rural areas and disaster zones. UAV-BSs can also be employed for deployment of temporary broadband and reliable communication network for scenarios such as football matches. Integrating UAV technology with advanced technologies such as Terahertz (THz) communication, Millimeter Wave (mmWave) transmission, and Reconfigurable Intelligent Surfaces (RIS) holds significant promise as a pioneering research avenue for attaining data-intensive applications with exceptionally high Quality of Service (QoS) [88]. Nonetheless, due to unpredictable dynamics such as UAVs' connectivity to mobile users and communication links, it becomes cumbersome to maintain accurate real-time tracking of data traffic for optimal data traffic and path management. To address this challenge effectively, the integration of AI approaches emerges as a solution that outperforms traditional offline optimization methods in terms of decision speed and adaptability to the ever-changing environment.

In the context of AI-based optimization techniques within UAV-assisted 6G networks, the dynamic nature of these networks imposes stringent requirements on various aspects of their operation. These requirements encompass accurate state sensing, real-time decision-making, and the precise execution of actions. Achieving these objectives presents significant technical challenges that need to be effectively addressed to harness the full potential of UAVs in 6G networks. One of the pivotal solutions is deployment of DTN to attain accurate information regarding traffic scheduling and path planning for task oriented and uninterrupted data flow between physical and virtual environment. This enables the virtual twin UAV to learn from the dynamic environment

and make accurate decisions to achieve effective traffic scheduling.

Li et al. [83] proposed a DT framework for accurate prediction of network states considering users' mobility in D2D environment. The authors focused on optimizing task offloading and energy consumption minimization of Mobile Edge Computing (MEC) system while taking into account various parameters including users' association, UAV trajectory, transmission power distribution, and allocation of computational resources. The authors utilized Double Deep Q-Network (DDQN) for users' association and determine UAV mobility patterns. Moreover, a closed form expression is developed for transmission power distribution and an iterative algorithm is deployed for allocation of computational resources. The obtained results demonstrate substantial minimization in energy consumption. Similarly, Liu et al. [84] also focused on energy conservation. The authors proposed DT enabled BS and UAV assisted Mobile Edge Computing (MEC) to render data services to the mobile user. The resource scheduling problem is formulated through MDP. Afterwards, DRL based Multi-Agent Proximal Policy Optimization (MAPPO) with attention scheme and Beta distribution (AB-MAPPO) algorithm is developed to obtain optimize task offloading policy. The authors performed extensive simulations and the obtained results demonstrate substantial improvement in energy conservation through the proposed scheme.

Furthermore, maintaining a balance between energy consumption and task completion within a specific time is critical to achieve a high level of task completion ratio, optimizing the sustainability of computing and communication systems in UAVs' resource-constrained environments. Therefore, Tang et al. [85] focused on improving task completion ratio in multi-UAV systems. The authors exploited the potential of DTN and employed DRL to improve efficiency and resource intensive utility of multi-UAV systems while focusing on dynamic task assignment. The proposed approach involves two stages of task assignment and task reassignment. Initially, tasks are partitioned into sub-tasks based on the shortest distance using a Genetic Algorithm (GA) and assigned to UAVs. Then, the DT is employed to enable UAVs learn from task features and generate the Q-value of the estimated value of DRL for UAVs via pre-training. Afterwards, the Q-value is applied to the Deep Q-Learning Network (DQN) in the UAVs that reduces the training episodes. Moreover, DQN is applied for training the task reassignment strategy. The obtained results show a remarkable improvement in task completion along with energy conservation. The aforementioned works highlight the energy constraint of UAVs, emphasizing the need to manage energy consumption efficiently. In this context, it is essential to focus on flight time because a longer flight duration not only improves the UAV's mission capabilities but also plays a pivotal role in conserving energy, ultimately ensuring the successful provision of data services. Thus, Soliman et al. [86] developed a smart navigation framework to minimize time and energy consumption. The

authors exploited DRL to learn users' mobility trajectory and plan energy efficient scanning strategies. A 3D physics-based DT is created for efficient exploration to train the DRL agent. Moreover, feasibility of the proposed mechanism is tested on a testbed based on an integration of Raspberry Pi and a Parrot Anafi drone. Extensive simulations reveal substantial gain in flight time and energy conservation in locating mobile users for effective data transmission.

IV. AI-DTN: APPLICATIONS

In this section, we delve into the implementation and functioning of 6G with a focus on the significant advantages rendered by DT. This section emphasizes the potential applications that will leverage 6G communication system capabilities. The requirements and key enabling technologies of each application are presented in Table 7.

A. SMART CITIES & URBAN PLANNING

The smart city market is projected to surpass USD 2376 billion by 2025 [89]. To become "smart," cities need people-centered technical innovations that improve the quality of life for all inhabitants, not just focusing on efficiency. However, conventional planning tools struggle to address the complexities of urban systems and the diverse needs of citizens. Digital technologies offer a substantial opportunity to bridge this gap and promote more intelligent and sustainable governance. By actively involving citizens and supporting democratic urban planning, these technologies provide a comprehensive perspective on the city. For instance, in the realm of virtual reality (VR), DT can play a crucial role in addressing the multifaceted challenges faced by smart cities. DTs encourage widespread citizen participation, foster collaboration, and facilitate consensus-building among various stakeholders, promoting innovative and collaborative approaches to urban planning [90], [91].

In the context of 6G DTNs, data serves as the core element, seamlessly flowing from the physical environment of smart cities to the corresponding virtual layer represented by DTs. Leveraging sensor technologies like IoT, cities acquire real-time, site-specific data to tackle emerging urban planning challenges related to urban morphology, weather events, climate conditions, accessibility, and proximity. Advanced techniques like AI and Edge computing process this data swiftly, providing valuable insights, patterns, and predictions. The combination of 6G DTNs and immersive technologies like VR and AR enables hyper-connectivity, creating a highly communicative environment for citizens and stakeholders to collaborate and optimize urban systems effectively. Furthermore, with an advent of geo-spatial "big data" has opened up new possibilities for hyper-connectivity between communities, the environment, and high-tech devices [92], [93]. Overall, 6G DTNs enable cities to harness the power of data and advanced technologies to create accurate virtual representations of their urban environments. These DTs serve as valuable tools for smart city and urban planners, administrators, and citizens to collaboratively address complex

TABLE 7. 6G application requirements and key enabling technologies.

Application	Description	Requirements				Key Enabling Technologies
		Data Rate	Latency	Spectrum Efficiency	Area Traffic Capacity	
Smart Cities	Data-driven urban development.	High	Low	High	High	Edge Computing, Terahertz Communications, Reconfigurable Intelligent Surface, Blockchain
Smart Industry	Smart manufacturing and automation.	High	Low	High	High	Edge Computing, Reconfigurable Intelligent Surfaces
Healthcare	Advanced healthcare communication.	High	Low	High	Medium	Edge Computing, Blockchain, UAV
IoT	Interconnected devices and sensors	High	Low	High	Medium	Edge Computing, Blockchain
Transportation and Logistics	Efficient and connected transportation.	High	Low	High	High	Terahertz Communications, UAV

challenges, develop smart solutions, and enhance the overall sustainability and livability of smart cities.

Dembski et al. [94] introduced a prototype of an urban DT designed for the town of Herrenberg, Germany. This prototype encompasses several key elements. Firstly, an urban DT model is configured through 3D modeling of environment. Secondly, street designing through graph theory is performed. Thirdly, the authors performed urban mobility and wind flow simulations. Finally, a collection of empirical quantitative and qualitative data obtained through volunteered geographic information (VGI). To enable accessibility and engagement, the urban DT is integrated into a VR visualization platform. The authors conducted a public demonstration of their prototype during various public participatory events, such as the Morgenstadt Werkstatt (Tomorrow's Cities Workshop). The obtained results reveal a significant role of the urban DT in facilitating participatory and collaborative processes for smart city planning. The DTN demonstrates its potential to empower citizens and stakeholders in decision-making, thereby enhancing the overall urban planning process.

Moreover, combining DT technology with AI algorithms can develop dynamic and real-time virtual representations that accurately reflect urban systems and interact with the physical world. This integration can enable data-driven decision-making, efficient resource management, and optimized traffic flow, leading to reduced congestion, lower fuel consumption, and improved air quality. Therefore, Salunke et al. [95] explored the potential of Reinforcement Learning (RL) enabled DTs for advancement in smart cities towards optimal urban dynamics. The integration of cutting-edge technologies, such as DTs and RL, empowers cities to create dynamic virtual representations that learn from real-world data and make intelligent decisions. The proposed approach enhanced traffic flow, which demonstrates the significant impact of RL-enabled DTs in reducing congestion, shortening commute times, and improving overall transportation efficiency in smart cities. Additionally, the

projected improvements in air quality, green spaces, and renewable energy adoption highlight the positive environmental implications of this approach.

Adreani et al. [96] proposed an automatic approach for generating 3D city models from diverse data sets and seamlessly integrating them into the open-source Smart City framework, Snap4City. Their solution enables the creation of comprehensive visualizations of 3D city elements combined with a wide range of Smart City data, including time-series data from IoT devices, heatmaps, traffic flows, bus routes, and cycling paths. The authors utilized a deep learning method based on a U-Net architecture for accurate rooftop detection and alignment. The implementation has been successfully incorporated into the Snap4City platform, allowing easy and intelligent navigation of the city's global DT and its associated data.

However, ensuring the security and trustworthiness of digital assets used in a DT is of utmost importance. Pervaz et al. [97] introduced a novel Smart city diGital twiN vERifiable Data (SIGNED) framework', which is built on principles of data ownership, selective disclosure, and verifiability. Through the use of Verifiable Credentials, SIGNED ensures the verifiable authenticity of DT data, encompassing aspects of provenance, transparency, and reliability through verifiable presentation. The evaluation of SIGNED demonstrates its success in facilitating the trusted and verifiable exchange of data at minimal performance cost, thereby enhancing security and addressing privacy concerns when sharing data among various functional units in a smart city.

B. SMART INDUSTRY

The integration of DTN into smart industry aligns with recent advancements in communication systems, opening possibilities for real-time interactions between DT and physical manufacturing systems. This integration facilitates

seamless information exchange and fostering closed-loop sensing and execution in manufacturing. An advent of 5G and its supporting technologies promises to underpin the convergence of DTN and AI within the manufacturing landscape in order to enable widespread connectivity of sensors, actuators, and controllers.

The management of the substantial volume of data in industrial and digital transformation domains necessitates the utilization of advanced architectures, techniques, tools, frameworks, and algorithms. For instance, Zhou et al. [98] introduced a big data processing framework tailored to industrial maintenance in a DT context. The authors proposed an Online Analysis DT (OADT) approach to perform real-time data analysis and control power grid operations for enhancing reliability and efficiency of energy distribution networks. The proposed scheme involves the seamless integration of the OADT with the existing power grid infrastructure for ensuring a smooth transition towards more intelligent and responsive grid management. The obtained results demonstrate that the proposed approach is efficient in reducing computation time. Many companies are proactively adopting the DT paradigm to explore the transformative potential of DT for manufacturing. For instance, Siemens utilized DT to reduce failure rates, accelerate time-to-market, and explore new business opportunities [99].

In the realm of smart industry, Cyber-Physical Systems (CPSs) emerged as foundational components of smart factories and production facilities. These systems bring together the physical and digital realms for creating a harmonious fusion of machinery, advanced technology, and data-driven intelligence. These smart machines are evolving to meet the highest standards of modern factory requirements, demonstrating the ability for bidirectional communication, efficient data handling, storage, analysis, and even exhibiting robust fault tolerance features. An integration of CPSs with DT emerges as a transformative pathway possessing profound implications for the advancement of smart manufacturing. The convergence of CPS and DT empowers manufacturers with improved efficiency, precision, and management enabling a seamless connection between vital components.

This transformative potential extends to dynamic scheduling in job-shop environments, exemplified by its application in optimizing milling machine operations for hydraulic valve production [100]. This dynamic scheduling hinges on DT's ability to leverage both physical and digital data to enable predictive analysis to ensure machine availability and facilitating prompt rescheduling. Moreover, a DT model for a small-scale knuckle boom crane is crafted for condition monitoring by incorporating nonlinear Finite Element (FE) analysis driven by factors such as payload weight. The model facilitates real-time determination of crucial parameters like strains, stresses, and load characteristics, significantly boosting safety and reliability. This approach holds promises for broader applications across various robotic manipulators in the industry.

DT models play a pivotal role in enhancing the functionality of HVAC systems, specifically in continuous anomaly detection within centrifugal pumps. This application streamlines operational procedures and facilitates more efficient asset management through automated asset monitoring [101]. The integration of data-driven DT models with advanced prediction techniques, such as Deep Stacked GRU (DSGRU) based on deep learning principles provides an opportunity to create predictive maintenance systems tailored for manufacturing machinery. To validate this concept, practical experimentation involved the utilization of vibration data from milling machine tools that effectively showcased the DT model's capability in predicting tool wear [102].

The aviation sector harnessed DT technology to create DTs for helicopter dynamic systems, with a specific focus on the swashplate rotor assembly. These DTs perform prediction of the lifetimes of mechanical components and the flight data is utilized to simulate the loads experienced by the mechanical components. The simulation models perform a critical role in designing and validating new models, which subsequently undergo bench trials to ensure their effectiveness and performance [103]. DT models also found practical applications in simulating the cutting process in Computer Numerical Control (CNC) machines. This application offers the potential for cost reductions, waste minimization, collision prevention, extended system life, and heightened accuracy and precision in the cutting process [104]. Moreover, where human operators and robots collaborate or share a workspace, DT models emerge as essential tools for enhancing safety, efficiency, and overall productivity. For instance, Malik et al. [105] developed a DT model to facilitate the design, construction, and control of human-machine collaboration within an industrial assembly setting. This approach effectively optimized human-robot cooperation, enhancing productivity and safety in complex smart industrial environments where humans and robots work together.

Security concerns are steadily increasing in the realm of smart industry. This has led to a thorough analysis of various cyber-attack modes within such systems that encompasses assessments of attack severity, categorization, and the safety of human workers involved in human-robot collaborations. To address these evolving threats, Khalid et al. [106] introduced a security framework for securing human-robot collaboration. Moreover, the authors proposed a defence mechanism to secure complex systems. The cyber-attacks are categorized based on their degree of controllability and potential impacts on CPS performance and efficiency, particularly focusing on human worker safety during human-robot collaboration. The proposed design follows a decentralized architecture for the Collaborative Robotic Cyber-Physical System (CRCPS) structure to eliminate single points of failure vulnerable to attacks, while taking into account the interdependent interactions between modules. The proposed security framework employs a dual-pronged strategy to

demonstrate its effectiveness through the NeCS-Car teleoperation benchmark and incorporates enhanced data security at crucial interconnected adapter nodes and an intelligent module. Chen et al. [107] also focused on security threats brought by DTN. The authors proposed the DT Protection Function (DTPF) and utilized three security schemes including cross-domain security classification, minimize privacy design, and multi-dimensional policy validation to ensure security.

Furthermore, 3D printers, as integrated digital devices, enable the design-as-manufacturing model and offer simplified manufacturing services for cloud manufacturing users. However, traditional cloud manufacturing models face challenges due to direct equipment access, including network strain, high latency, and security risks. Therefore, Guo et al. [108] focused on enabling the cloud-based access of manufacturing equipment. The authors proposed a cloud-edge collaboration architecture based on DT. The proposed architecture extends cloud services to the edge for enhancing network reliability. The proposed approach showcases the edge platform's capabilities to verify the effectiveness of the DT-based 3D printer cloud-edge collaboration system for providing technical support for future cloud manufacturing implementations.

C. HEALTHCARE

The utilization of information and communication technology (ICT) in healthcare has surged due to the impact of COVID-19. Ensuring uninterrupted healthcare services has triggered both business and research endeavors aimed at digitalizing the procedures required for delivering health services, particularly during emergencies. Digital collection, transmission, and processing of health data are at the core of this innovation process.

In the rapidly approaching era of 6G networks, the healthcare landscape is poised for profound transformation. Embracing the potential of 6G DTNs, healthcare delivery and medical practices are on the cusp of radical change. By harnessing the capabilities of 6G DTNs to create dynamic virtual representations of real healthcare systems, the future holds the promise of personalized, data-driven medicine, seamless remote patient monitoring, and enhanced decision-making for healthcare professionals. As the world prepares for the dawn of the next generation of connectivity, the convergence of 6G and DTNs in healthcare ushers in a new era of intelligent, efficient, and accessible healthcare solutions. With the anticipated prominence of digital health in 6G wireless cellular networks, the scope for utilizing DTs in health and wellness applications has expanded to include living and non-living entities. Laamarti et al. [109] proposed ISO/IEEE 11073 standardized DT framework for health services. The proposed framework encompasses the processes of data collection from personal health devices, data analysis, and rendering feedback to users in a continuous loop cycle. Additionally, the proposed framework interfaces

non-X73 compliant health devices with X73 wrapper module that allows an incorporation of both X73 compatible and non-X73 compliant health devices. Moreover, a flexible X73 mobile application is designed that can be utilized with any X73 compatible device. The experimentation demonstrate that the proposed mechanism effectively gathers individuals health data and provides feedback to the individuals.

Firouzi et al. [110] introduced Personal DTs (PDTs) to replicate human organs and offer comprehensive health information. PDTs merge data from electronic health records, clinical data, wearables, smartphones, and social media to provide a 360-degree view of an individual's health. PDTs can alert users and healthcare professionals to patterns indicating potential health concerns of users. When combined with AI, PDTs become powerful tools for predicting user contexts, detecting early warning signs, forecasting changes, and enabling personalized medicine. Hence, PDTs have the potential to revolutionize preventive healthcare and personalized medicine. Furthermore, for cardiac issues, Elayan et al. [111] introduced an AI-based classifier model for electrocardiogram (ECG) heart rhythms. The proposed model utilizes AI-enabled IoT and data analysis for accurate predictions of heart conditions. The obtained results demonstrate that integrating DT with the healthcare sector can lead to improved healthcare processes and creation of a smart and scalable health ecosystem to foster collaboration between patients and healthcare professionals.

D. TRANSPORTATION AND LOGISTICS

In the realm of transportation, the widespread adoption of Intelligent Transportation System (ITS) technology has led to the extensive deployment of sensors and connectivity throughout the transportation ecosystem. The development of ITS holds significant potential and capabilities for future smart cities, contributing to the establishment of intelligent, efficient, and secure transportation systems. As the complexity of interactions and the volume of traffic data generated by ITS increase, managing and controlling these systems becomes more challenging [112]. Consequently, the integration of the extensive data from ITS with the computational, communication, and control (3C) capacities of potential ITS managers can enhance the quality, stability, and reliability of ITS services [113].

To infuse intelligence into the digital representation of physical systems within the cyberspace context, the concept of DT can be harnessed to enhance ITS services. The advantages of employing DT-related technologies in ITS primarily stem from comprehensive life cycle support, real-time accurate responses, and bidirectional communication. Employing DT for ITS enables cohesive scheduling and management, mapping intricate and dynamic intelligent transportation subsystems, such as private vehicles, public transit, and energy systems. By leveraging physical models and sensory data, DT can continually forecast the system's health, lifespan, and critical security events [19]. DT effectively

gathers extensive sensory data from ITS to offer a holistic perspective, while ITS benefits from the predictive power of DT's robust computation, communication, and control capabilities, aiding in shaping the system's future [114]. Various applications have been developed to demonstrate the utility of DT in ITS systems, including cooperative ramp merging [115], cooperative driving [116], and driver behavior modeling [117], [118]. These applications showcase how DT enhances ITS by enhancing safety and mobility in existing transportation frameworks.

Gonzalez et al. [119] proposed a DT of a vertical transportation system, focusing on how to model this DT, and using it to evaluate the system condition and potential corrective solutions. Exploiting the advantages of object-oriented modeling, this DT has been built ensuring that it may be dynamically adapted to different monitoring scenarios. Moreover, its adaptability is tested showing that reduced models behave similarly. The resulting reduced models are used to estimate installation parameters in a bottom-up way, using the measurements of a scaled test bench. The estimated parameters are used to update a high-fidelity model and simulate the effect of corrective actions.

In the context of Intelligent Transportation Systems (ITS) in smart cities, DT technology assumes a pivotal role by serving as a foundational and fundamental technology to ensure reliable services for a diverse array of DT beneficiaries, including private vehicles, public transit, and energy systems. The collaboration between DT and ITS holds the potential to unlock the capabilities of smart city participants, enhancing operational efficiency and dependability. However, the challenge of achieving a secure and dependable alignment between DT and ITS remains critical. Addressing this challenge, Liao et al. [120] introduced a blockchain-driven DT as a Service (DTaaS) mechanism that optimally leverages the inherent sensing capacities of ITS while synergizing them with DT's broader perspective. Moreover, a dual-action model and price adjustment algorithm are designed to achieve optimal DT matching for ITS requesters while safeguarding the interests of all participants. Furthermore, through the implementation of a permissioned blockchain and the innovative DT Delegated Proof of Stake (DT-DPoS) consensus scheme, information confidentiality is upheld. Empirical results underscore the efficacy of this proposed approach in efficiently stimulating and facilitating DT transactions.

The coexistence of connected and automated vehicles (CAVs) alongside human-driven vehicles (HDVs) on roadways is predicted in future. Consequently, adopting an approach that considers the mixed traffic environment becomes more pragmatic, given that the smooth operation of CAVs can be disrupted by interactions with HDVs. Therefore, it is essential to develop CAVs mechanisms to perform actions while taking into account evolving human driving behaviors. Liao et al. [121] introduced Driver DT (DDT) for real-time prediction of personalized lane change behaviors. The proposed approach empowers CAVs to

anticipate the actions of surrounding vehicles with the aid of DT Technology (DTT). DTT is deployed within a vehicle-edge-cloud architecture, where cloud servers model driver behavior for individual HDVs based on historical driving patterns. Then, edge servers process real-time data from each driver, leveraging their respective DTs on the cloud to predict lane change maneuvers. The prediction process involves utilizing a sequence-to-sequence LSTM neural network to determine lane change patterns. Additionally, distinct driving behaviors are modeled for various drivers, where preferences are obtained and analyzed through inverse reinforcement learning based on historical data in the cloud server. It is evident from the obtained results that the proposed approach can predict target driver's lane change intention within 6.08 seconds.

Dasgupta et al. [122] proposed DT for Adaptive Traffic Signal Control (ATSC) to optimize users' driving experience by minimizing waiting times at intersections and signals. However, it is challenging to address traffic congestion delays. Therefore, the authors proposed a DT based ATSC (DT-based ATSC) scheme to predict vehicles' waiting times in congestion conditions. The authors modeled a digital clone of an urban roadway environment. This environment facilitated the collection of data related to vehicles, traffic signal, and traffic patterns. The obtained results demonstrate that the proposed scheme efficiently reduces vehicles' cumulative waiting time and optimizes driving experience.

E. INTERNET OF THINGS

In the landscape of 6G communications, the convergence of DTNs and AI with IoT applications emerges as a transformative paradigm to bring forth a new era of intelligent, context-aware systems. One of the key aspects of this integration is the enhancement of predictive analytics. AI algorithms enhance DTs by predicting and analyzing behaviors, anomalies, and trends through their real-time data representation [123], [124]. This predictive capability is particularly vital in IoT applications where preemptive decision-making is crucial, such as in healthcare for predicting patient conditions or in smart cities for optimizing traffic flow. The adaptive learning capabilities of IoT devices can be further enriched through the integration of AI and DTN. AI algorithms embedded in DTs can dynamically adjust and optimize the behavior of IoT devices based on changing environmental conditions or user preferences. This adaptability ensures that IoT applications can continuously evolve and improve their performance. Guo et al. [125] proposed a DT framework for Device-to-Device (D2D) communications within the 6G IoT environment. The objective is to optimize communication links for IoT end devices by leveraging edge networks, utilizing Federated Learning (FL) and Reinforcement Learning (RL) for decentralized resource allocation. The implementation of a DT-powered learning mechanism ensures privacy awareness in resource allocation,

highlighting the substantial potential of DT in optimizing communication links for IoT devices in the field of IoT.

Expanding on the synergy of AI and DT technology for IoT applications in 6G, Xu et al. [126] delved into the intricate interplay between cyber systems and physical industrial entities. They introduce an AI-enabled cyber-twin architecture, leveraging Deep Reinforcement Learning (DRL) for systematic trial and error within the cyber-twin model. The Adaptive Observation Window for Deep Q-Network (AOW-DQN) is introduced to adapt system states to the dynamic nature of cyber-subsystems such as control, communication, and computing (3C). The authors utilized robot arm positioning control to determine efficiency of the proposed mechanism. The obtained results demonstrate that AOW-DQN achieved lowest error between robot arm's defined and current positions. This highlights the effectiveness of AI-enabled architectures in enhancing the precision and adaptability of industrial IoT systems within the 6G.

Industrial IoT (IIoT) is a transformative force in the manufacturing sector, leveraging connectivity, data analytics, and intelligent systems to optimize processes and enhance overall efficiency. In the context of 6G, the integration of DT technology and AI further propels the capabilities of IIoT, enabling real-time monitoring, predictive maintenance, and adaptive decision-making in industrial environments. Therefore, Yang et al. [127] harnessed DT technology within the realm of industrial IoT to actively capture real-time information from devices. This approach enabled the system to perform intelligent decisions while taking into account the inherent heterogeneity among various IoT devices. The authors employed FL to ensure data privacy and reduce information transmission burden. To overcome the challenge of IoT devices diverse characteristics, the authors utilized DRL to select appropriate devices within FL. This integration of DT technology, FL, and DRL not only enhances the real-time capabilities of Industrial IoT but also ensures a more streamlined and intelligent decision-making process, ultimately contributing to the optimization of the overall system performance. The obtained results demonstrate a significant performance gain in terms of convergence rate and accuracy.

Security and privacy in IoT applications are also bolstered through the integration of AI and DTN. When DTs are equipped with AI-driven security measures, then in real-time potential threats or vulnerabilities can be detected and responded. This proactive approach enhances the resilience of IoT networks against cyber-attacks and ensures the privacy of sensitive data. Islam and Shin [128] proposed a DT based data collection approach employing FL through drone, where the data is secured through blockchain. Firstly, the authors integrate differential privacy into the system to enhance data privacy before any information is shared. This mechanism ensures that individual data contributions remain confidential even during the collaborative learning process. Secondly, the authors introduce a multi-role training mechanism to adapt and optimize its training. This enhances the efficiency of the

learning model and ensures that each participant contributes optimally to the FL. Furthermore, the implementation features a duplex design verification architecture leveraging performance checks and the Hampel filter. This architecture serves as a robust means of validating the integrity and accuracy of the system. Performance checks enable real-time monitoring and evaluation of the system's efficiency, while the Hampel filter acts as a robust statistical tool to identify and mitigate outliers or anomalies in the data, contributing to the overall reliability of the system. Finally, an authentication technique is introduced for combining time frame checks and the cuckoo filter. This technique adds an additional layer of security by verifying the temporal consistency of data contributions, ensuring that information is up-to-date and preventing potential manipulation or tampering. The cuckoo filter, known for its efficiency in set membership testing, further enhances the system's ability to authenticate and validate incoming data. Collectively, these measures constitute a comprehensive strategy to fortify the privacy, reliability, and security of the proposed system. The obtained results demonstrate that the proposed approach substantially improved the performance in terms of block circulation rate, loss, and time.

V. LESSONS LEARNED, CHALLENGES AND FUTURE RESEARCH DIRECTIONS

In the previous sections, we provided a detailed discussion of DTN for 6G. Then, we highlighted fundamental techniques of AI 6G DTN. Afterwards, we discussed enabling technologies and application of AI 6G DTN. We performed an extended survey of AI 6G DTN and discovered key current issues that will shape the research community's future orientations. To enable AI 6G DTN, it is imperative to overcome these challenges and design optimal DTNs for 6G networks. In this section, we discuss lessons learned and highlight existing challenges and future research directions that will pave the way for the readers to conduct research. In Table 8, we provide a description and outline future research direction for existing challenges.

A. LESSONS LEARNED

The communication system services are experiencing rapid evolution. This evolution necessitates the development of flexible DTNs that can easily adapt to new services. Furthermore, AI can be employed to train DTNs for adaptability across various scenarios. This approach significantly reduces the time and cost associated with DTN design. Looking ahead, 6G is poised to become an integral part of every service and application that relies on wireless resources and user interaction with DTN virtual objects. To ensure efficient DTN deployment within communication systems, whether in cloud or edge servers, a co-design approach involving both software and hardware is critical. This entails careful consideration of cost implications and the precise mathematical modeling of the physical environment. AI a data-driven model, can facilitate real-time adjustments and precision, ensuring the

network's adaptability and high performance within the 6G landscape [129].

The integration of AI and DTNs can be viewed as the backbone of 6G. However, it is essential to acknowledge that despite their immense potential, these advanced technologies introduce inherent vulnerabilities and security concerns. Moreover, the advent of 6G brings forth a multitude of services and applications, thereby presenting unforeseen challenges such as distributed denial of service attacks, physical infrastructure vulnerabilities, and regulatory complexities. These challenges demand continuous attention and the development of innovative solutions to fully unlock the potential of 6G networks. Efficient computing resources and algorithms for digital twinning are a necessity [130]. In this context, data-driven decision-making, harnessing the potential of AI-enabled DTN, plays a fundamental role in addressing these challenges. It enables informed responses to security threats and vulnerabilities, serving as a cornerstone in the comprehensive strategy required to secure and optimize 6G networks for the future.

Furthermore, in the realm of 6G, AI-enabled DTN assume a pivotal role in the transformation of Industry 4.0. As the foundation for the next industrial revolution, Industry 4.0 places a strong emphasis on connectivity, automation, and data-driven decision-making. The deployment of DT in proximity to smart factories at the network edge level can significantly enhance data reliability and reduce latency, allowing real-time monitoring and control of industrial processes. For instance, the utilization of data analytics and FL for computations can bolster data security, reliability, and privacy in smart factories [131]. FL can facilitate collaborative model training without sharing sensitive data externally, ensuring data privacy while enabling machine learning models to continuously improve.

The convergence of AI-enabled DTN within the smart industrial paradigm results in more accurate predictions, computations, and decision-making. AI-powered quality control systems can detect defects in real-time, ensuring product quality. Moreover, AI algorithms can optimize supply chain logistics, inventory management, and energy consumption, thereby increasing operational efficiency and sustainability. Hence, the integration of AI and DTN in 6G not only revolutionizes processes but also empowers industries to remain competitive in an increasingly digital and interconnected world. The deployment of AI-enabled DTN can pave the way for a smarter, more connected, and data-enriched future, where the limitless potential of AI and DTN can be harnessed to drive progress and innovation across various sectors.

B. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

The deployment of AI-enabled DTN in 6G introduces several challenges that require advance approaches to harness transformative technology. These challenges can influence deployment of DTN in terms of efficiency and adaptability.

In this subsection, we delve into the existing challenges and future research directions.

1) MOBILITY

Users' mobility is inherently dynamic and unpredictable. Users' mobility is erratic and influenced by many factors such as traffic patterns, events, and users' behavior. This poses a significant challenge in accurately modeling DT networks for mobility scenarios. In response to these challenges, AI technologies emerge as indispensable tools for reshaping the future of mobility in 6G DTNs. For instance, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) can be employed to capture patterns and temporal dependencies in users' movements [132]. Historic mobility data and real-time contextual information can be utilized to predict users' mobility trajectories and locations. The predictive modeling empowered by AI enables DTN to optimize real-time resource allocation for users and network planning.

Moreover, AI-driven Predictive Maintenance (PdM) approaches provide a transformative solution to enhance the efficiency and reliability of mobility services such as intelligent traffic management systems in 6G DTNs. PdM leverages data analytics and AI for prediction and prevention of equipment failures, reducing maintenance costs, and minimizing downtime. For instance, AI algorithms can reduce congestion, improve users' mobility, and traffic flow by leveraging real-time data and information of connected vehicles for decision-making. This proactive technique can substantially reduce network operations disruptions and improve network services. However, AI-driven mobility models depend on data collection including users' personal information [133]. It necessitates safeguarding data security breaches and privacy infringements. This demands development of robust security schemes and compliance with varying data protection laws.

One prominent approach lies in exploration of RL with DTNs to enable autonomous decision-making within DTN. This approach can learn and adapt to evolving mobility scenarios by optimizing resource allocation and network planning. The integration of AI and DTN paves the way for agile adaptive networks that can accommodate evolving users' mobility trajectories. Recently, Transfer Learning (TL) has been employed in DT migration as a means to save time and network resources [31]. However, implementing TL within DTNs presents its own set of challenges. The presence of heterogeneous hardware/software configurations among different edge servers may pose challenges to the seamless application of TL in the DTN. Transferring a AI model from one server to another may encounter interoperability issues. Additionally, the careful selection of an appropriate new base station (BS) or edge server to meet the evolving mobility demands is a complex task. To achieve optimal performance of DTN, it is essential to consider factors such as service time for the physical object, incentives,

network conditions, mobility and operation states, and the availability of various BSs is essential [134]. These concerns needs further investigations. These complexities underline the importance of further investigations and in-depth research in the dynamic intersection of AI and DTN to effectively address mobility challenges and propel the evolution of these transformative networks.

2) SECURITY & PRIVACY

The use of AI-enabled DTN in 6G will face a range of security and privacy issues due to its complex and interconnected nature. DTNs are designed to replicate real-time physical objects and environments into virtual environment, therefore, DTNs need to handle vast amounts of sensitive data and facilitate critical interactions between physical and digital entities. This causes DTNs susceptibility to various threats and vulnerabilities that can compromise the integrity, confidentiality, and availability of information. One of the primary security concerns is the risk of data breaches. DTNs need to store and process sensitive information related to physical assets, and user behaviors [135]. If unauthorized parties gain access to these crucial data through cyber-attacks, it can lead to serious issues such as intellectual property theft, operational disruptions, and potential safety hazards. Moreover, identity and access management is critical in DTNs to prevent unauthorized access. Hence, efficient authentication mechanisms are necessary to ensure that only legitimate users can interact with the DTN system. However, without robust identity management, malicious actors may gain unauthorized access, leading to data manipulation, service disruptions, or unauthorized control of physical assets. AI brings opportunities for enhanced threat detection and anomaly detection in DTNs. Developing AI-driven security mechanisms for real-time threat identification and mitigation is an ongoing research area. AI-based intrusion detection systems and behavior analytics can help in identifying unusual activities that may indicate security breaches [136].

In addition, privacy is also a serious concern in AI-enabled DTNs during handling personal or sensitive information. Compliance with data protection regulations is essential to safeguard users' privacy and prevent the unauthorized use of personal data [137]. To address these security and privacy challenges, blockchain's decentralized, immutable, and transparent characteristics can be devised for data integrity and privacy within the context of 6G DTN powered by AI. Each data transaction within the AI-driven DTN ecosystem can be recorded in a block and linked through cryptographic hashes, ensuring immutability and transparency. The distributed nature of blockchain reduces the risk of single points of failure and enhances resilience against cyber-attacks. Furthermore, blockchain's consensus mechanism ensures agreement among network participants, validating the accuracy of data and preventing unauthorized changes in the data. This collaborative approach to securing

data in AI-driven DTNs enhances trust and confidence among users and stakeholders.

Moreover, blockchain can also be employed to achieve privacy in DTNs. It enables users to control the sharing of their data through permissions and access controls, which ensures that sensitive information remains private and accessible only to authorized parties [138]. The integration of AI within DTNs further enhances the potential for fine-grained, context-aware data access and privacy management, aligning data sharing with individual preferences and consent. While blockchain offers promising solutions to enhance security and privacy in AI-driven DTNs, there are challenges to consider such as scalability and performance issues. Research and innovation efforts are needed to optimize blockchain's scalability to handle the high transactional demands of real-time DT systems without compromising performance. Balancing the computational demands of AI and the stringent privacy and security requirements of DTNs represents a key area of exploration for future research and development initiatives.

3) 6G DTN STANDARDIZATION

In DTNs, interoperability among diverse DT platforms is of paramount importance. It requires defining common data formats, communication interfaces, and protocols for seamless data exchange and cooperation among various DTNs. AI can aid in defining common data formats and communication interfaces [139]. Standardizing AI algorithms and models for data exchange between DTs can foster seamless interoperability, ensuring that diverse DT platforms can communicate and collaborate effectively. Moreover, robust standardization of authentication protocols, encryption, and privacy ensuring approaches is necessary to maintain information secrecy and privacy. DTNs will experience a large volume of data, necessitating the standardization of scalable architectures, distributed processing, and data storage mechanisms. However, DTNs require accurate and dynamic representation of physical entities that demand low latency processing and high data streaming for responsive interactions between physical and virtual environments. For instance, in an integration of edge and cloud computing resources in 6G DTNs require standardizing interfaces and protocols to facilitate seamless integration, data offloading, and synchronization between edge and cloud infrastructure. AI can contribute to the development of advanced security measures, such as AI-driven threat detection and privacy-preserving techniques, to enhance the security of data exchange within DTNs. For instance, federated DTs can be devised for collaboration and data sharing among distributed DTNs [140].

The standardization of AI in 6G DTNs can unlock the potential for advanced and real-time analytics. This can ensure seamless interoperability and efficiency across diverse DT platforms. This requires common AI algorithms, models, and interfaces for seamless communication and data

TABLE 8. Summary of key research challenges and directions.

Challenges	Description	Future research directions
Mobility	- Users' mobility is highly dynamic, and AI can be employed to accurately model users' mobility trajectories. This can enable DTN to accurately determine users' future mobility patterns and optimize resource allocation and network planning. - Dynamic resource allocation strategies that consider mobility patterns and operation states can enhance the performance and reliability of DT networks.	- AI needs to be explored to improve precise predictions of users' locations by incorporating various contextual factors, such as traffic conditions, events, and user behaviors. - AI can be used in DTN for dynamic server selection considering factors including service time, network conditions, and incentives in mobility environment.
Security	- Blockchain's decentralized and transparent nature makes it suitable for ensuring data integrity. Each data transaction can be recorded in a blockchain, creating an immutable and transparent ledger. - Data security can be achieved through block chain's permission and access control feature to maintain privacy of sensitive information.	- It is crucial to develop automated tools for data anonymization and users' consent management. - development of privacy-preserving smart contracts and access control mechanisms tailored to the needs of DTNs. - It is necessary to develop users' mobility aware robust security measures, including encryption protocols, access controls, and authentication mechanisms, to protect data integrity and confidentiality in mobile DT environments. - Investigate methods for secure and privacy-conscious data sharing among mobile DT nodes. - Develop strategies for real-time threat detection and response to safeguard against malicious activities targeting mobile DT networks.
6G DTN Standardization	- Interoperability among diverse DT platforms is imperative that requires establishment of common communication interfaces, protocols, and data formats to achieve seamless data sharing and cooperation. - Standardizing AI-enabled 6G DTN is necessary for massive amount of data processing, effective data handling, architecture scalability, distributed processing, and data storage.	- Develop robust standardization for authentication protocols, encryption, and privacy-preserving techniques to maintain data confidentiality and privacy across DTNs. - Develop standards for high-speed data streaming and data synchronization between DTs and the physical world. - Investigate the use of federated DTs for collaboration and data sharing among distributed DT networks. - Investigate AI optimization and compression techniques to improve performance in resource-constrained DTN environments.
Data Availability & Exchange-Ability	- Seamless availability and exchange of data among diverse components. - Critical for continuous, accurate, and updated depictions of physical objects in DTNs.	- Developing innovative solutions for handling diverse data formats and communication protocols. - Developing robust data encryption methods, access controls, and anonymization approaches. - Investigating edge computing solutions and low-latency communication protocols.
Precision Modelling & Energy Conservation	- Precise modeling of the physical environment is essential for efficient network management, task offloading, and resource allocation. 6G DTN process large and high quality data that requires development of advance approaches for optimal training and decisions while avoiding redundant learning tasks for energy conservation.	- Exploring technologies such as THz and RIS to improve capacity. - Investigate periodic and event-triggered data collection approaches for optimal resource utilization

exchange between DTs. AI in 6G DTN standardization will focus optimization and compression to improve performance of resource constrained environments. A large number of users and devices will communicate with each other and there will be mobile devices as well that influences networks

performance. For optimal DTN performance, AI can be devised for accurate modeling of users' mobility is crucial. Hence, standardization of AI algorithms for adaptive learning and decision making is required to enable dynamic adjustments based on evolving network conditions.

4) DATA AVAILABILITY & EXCHANGEABILITY

In 6G, the integration of DTN and AI holds the promise of enhancing real-time and dynamic representations of the physical environment. This integration relies on the seamless availability and exchange of data among diverse components, including sensors, IoT devices, and cloud platforms. However, challenges emerge due to the diverse nature of data formats, structures, and protocols, necessitating solutions for uniform data exchange throughout the network. In addition, the interoperability across various platforms in DTN further increases the challenges that demand the establishment of shared data semantics and consistent data interpretation. For instance, a scenario where sensor data from different manufacturers employs distinct data formats and communication protocols, such as JSON and XML [141]. Here, ensuring seamless exchangeability requires the development of middleware or translation layers capable of harmonizing data formats and protocols for coherent interpretation across the network.

Real-time synchronization of data is critical for DTNs to ensure a continuous, accurate, and updated depiction of physical objects. Delays in data synchronization can compromise the effectiveness of representations, leading to inefficient decision-making. For instance, an industrial DTN may aim to synchronize real-time data from various IoT devices and machines to monitor and optimize production processes. Delays in synchronizing critical parameters, like temperature or pressure, could result in sub-optimal decision-making [142]. Implementing efficient edge computing solutions and low-latency communication protocols becomes essential to ensure timely synchronization and maintain the accuracy of the DTs.

Additionally, the exchange of data between different components introduces challenges related to data privacy, reliability, and security, especially concerning sensitive information. Robust data encryption, access controls, and data anonymization approaches are imperative to address these concerns and mitigate unauthorized access or data breaches. Concerning data privacy and security challenges, imagine a healthcare application of a DTN where patient monitoring devices continuously exchange sensitive health data. Robust encryption methods, access controls, and data anonymization techniques are imperative to ensure data privacy and security. Compliance with stringent healthcare regulations, such as HIPAA, further emphasizes the need for robust security measures to protect patient identities and mitigate unauthorized access [143].

The integration of AI in DTNs enhances adaptability and decision-making capabilities. AI algorithms embedded in DTs dynamically adjust and optimize the behavior of devices based on changing environmental conditions or user preferences. This adaptability ensures that applications can continuously evolve and improve their performance. The robustness of these AI-enhanced DTNs relies on continuous innovation and the development of scalable approaches to handle the increasing volume of data without compromising

system performance, latency, and the seamless exchange of data. This holistic approach aligns with the demands of 6G and necessitates AI-driven advancements in the realm of DTNs.

5) PRECISION MODELLING & ENERGY CONSERVATION

In 6G DTNs, precise DT modeling is crucial to provide efficient services such as network management, task offloading, and resource allocation. However, there are communication and computation constraints hindering precise modeling of physical environments. Therefore, real-time processing of physical object information necessitates the utilization of advanced technologies such as THz communications, RIS, and AI integration into wireless systems for ensuring ultra-high capacity and reliability communications. The integration of AI into DTN can play a pivotal role in addressing these constraints by leveraging AI and predictive analytics to enhance the precision and efficiency of DT modeling. Additionally, there is a dire need to develop lightweight and efficient modeling approaches to overcome computation resource limitations [144]. AI techniques including lightweight machine learning models and edge AI, can significantly contribute to addressing this challenge by reducing the computational overhead and enabling real-time modeling and decision-making with minimal resource consumption.

Moreover, 6G DTN heavily depends on large and high quality data that require energy efficient mechanisms for model training and decision makings. Therefore, data-driven models need to prioritize minimal data collection while ensuring the accuracy and reliability of DT tasks [145]. In addition, periodic and event-triggered data collection approaches can be employed for optimal resource utility. The Periodic data collection enables DTN to request the network for the desired data in a periodic manner, where the periods are adjusted based on the cost of data collection. Then, the event-triggered approach transmits data during specific events. It is essential to develop adaptive data collection methods, which can be further enhanced with AI-driven techniques, in order to control the volume and frequency without compromising performance.

There is a need to deploy AI approaches with lower power consumption to optimize predictions for maximizing energy conservation. Recently, FL based DT modeling is explored in 6G DTNs [146]. FL has the ability to train the model locally and transmit local model parameters to the server for aggregation. This approach can provide high level performance gain in the data transmission volume. However, incentivizing physical objects and servers to actively contribute their resources to DT modeling remains a challenge. AI can play a crucial role in optimizing FL-based modeling and resource allocation, ensuring that AI models are energy-efficient while achieving the desired precision.

It is necessary to achieve a balance between modeling precision and cost while taking into account selection

of participating devices with varying communication and computation times [147]. AI-driven optimization algorithms can assist in achieving this balance by dynamically adjusting the participation of devices based on their capabilities and resource availability. Looking at the potentials of AI in widespread applications across various fields such as medical, space technology, and manufacturing, AI extensive exploration in 6G DTN is imperative to foster trust-based joint learning and avoid redundant learning tasks across domains within the DTN. AI's ability to facilitate cross-domain learning and knowledge transfer is essential for achieving efficiency and synergy in 6G DTNs.

VI. CONCLUSION

In our survey, we have undertaken a comprehensive exploration of the integration of AI and DTN within the 6G communication system. Our approach involved the establishment of a taxonomy, delineating the complex landscape of AI-enabled DTN and associated services within the realm of 6G. Additionally, we delved into the critical sphere of key enabling technologies that play a pivotal role in the development of DTN, shedding light on the fundamental attributes and characteristics that define DTN in the context of 6G. Moreover, we highlight the existing challenges and potential avenues of future research. Our objective is to furnish readers with a profound understanding of the seamless integration of AI and DTN and the instrumental utilization of key enabling technologies to obtain optimal functioning of the 6G communication system. We expect that our survey will provide a comprehensive resource for providing a clear understanding and motivation for researchers for further exploration and innovation in the evolving landscape of AI-enabled DTN in 6G.

REFERENCES

- [1] J. De Vriendt, P. Laine, C. Lerouge, and X. Xu, "Mobile network evolution: A revolution on the move," *IEEE Commun. Mag.*, vol. 40, no. 4, pp. 104–111, Apr. 2002.
- [2] E. Dahlman, B. Gudmundson, M. Nilsson, and A. Skold, "UMTS/IMT-2000 based on wideband CDMA," *IEEE Commun. Mag.*, vol. 36, no. 9, pp. 70–80, Sep. 1998.
- [3] J. Gozalvez, "First commercial LTE network [mobile radio]," *IEEE Veh. Technol. Mag.*, vol. 5, no. 2, pp. 8–16, Jun. 2010.
- [4] D. Astely, E. Dahlman, A. Furuskär, Y. Jading, M. Lindström, and S. Parkvall, "LTE: The evolution of mobile broadband," *IEEE Commun. Mag.*, vol. 47, no. 4, pp. 44–51, Apr. 2009.
- [5] K. Shafique, B. A. Khawaja, F. Sabir, S. Qazi, and M. Mustaqim, "Internet of Things (IoT) for next-generation smart systems: A review of current challenges, future trends and prospects for emerging 5G-IoT scenarios," *IEEE Access*, vol. 8, pp. 23022–23040, 2020.
- [6] R. Piqueras Jover and V. Marojevic, "Security and protocol exploit analysis of the 5G specifications," *IEEE Access*, vol. 7, pp. 24956–24963, 2019.
- [7] M. A. Uusitalo, P. Rugeland, M. R. Boldi, E. C. Strinati, P. Demestichas, M. Ericson, G. P. Fettweis, M. C. Filippou, A. Gati, M.-H. Hamon, M. Hoffmann, M. Latva-Aho, A. Pärssinen, B. Richerzhagen, H. Schotten, T. Svensson, G. Wikström, H. Wymeersch, V. Ziegler, and Y. Zou, "6G vision, value, use cases and technologies from European 6G flagship project Hexa-X," *IEEE Access*, vol. 9, pp. 160004–160020, 2021.
- [8] W. Jiang, B. Han, M. A. Habibi, and H. D. Schotten, "The road towards 6G: A comprehensive survey," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 334–366, 2021.
- [9] C.-X. Wang, X. You, X. Gao, X. Zhu, Z. Li, C. Zhang, H. Wang, Y. Huang, Y. Chen, H. Haas, J. S. Thompson, E. G. Larsson, M. D. Renzo, W. Tong, P. Zhu, X. Shen, H. V. Poor, and L. Hanzo, "On the road to 6G: Visions, requirements, key technologies and testbeds," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 905–974, 2nd Quart., 2023.
- [10] J. F. Monserrat, D. Martin-Sacristan, F. Bouchmal, O. Carrasco, J. Flores de Valgas, and N. Cardona, "Key technologies for the advent of the 6G," in *Proc. IEEE Wireless Commun. Netw. Conf. Workshops (WCNCW)*, Apr. 2020, pp. 1–6.
- [11] F. H. P. Fitzek and P. Seeling, "Why we should not talk about 6G," 2020, *arXiv:2003.02079*.
- [12] P. Doorn, I. Dillo, P. Witkamp, D. S. Katre, and D. L. Giarretta, "Building a federated infrastructure for preservation of and access to research data in The Netherlands: The front office-back office model," in *Proc. APA/C-DAC Int. Conf. Digit. Preservation Develop. Trusted Digit. Repositories*, 2014, pp. 72–77. [Online]. Available: <https://api.semanticscholar.org/CorpusID:113455428>
- [13] M. Latva-Aho and K. Leppänen, "Key drivers and research challenges for 6G ubiquitous wireless intelligence," 6G Flagship, Univ. Oulu, Oulu, Finland, Tech. Rep., 2019. [Online]. Available: <http://urn.fi/urn:isbn:9789526223544>
- [14] Y. Wu, K. Zhang, and Y. Zhang, "Digital twin networks: A survey," *IEEE Internet Things J.*, vol. 8, no. 18, pp. 13789–13804, Sep. 2021.
- [15] D. E. Jones, C. Snider, A. Nassehi, J. M. Yon, and B. J. Hicks, "Characterising the digital twin: A systematic literature review," *CIRP J. Manuf. Sci. Technol.*, vol. 2, pp. 36–52, May 2020. [Online]. Available: <https://api.semanticscholar.org/CorpusID:216362509>
- [16] X. Lin, L. Kundu, C. Dick, E. Obiodu, T. Mostak, and M. Flaxman, "6G digital twin networks: From theory to practice," *IEEE Commun. Mag.*, vol. 67, no. 1, pp. 72–78, Nov. 2023.
- [17] R. Pillai and H. Babbar, "Digital twin for edge computing in smart vehicular systems," in *Proc. Int. Conf. Advancement Comput. Comput. Technol. (INCACT)*, May 2023, pp. 1–5.
- [18] Q. Qi, F. Tao, T. Hu, N. Anwer, A. Liu, Y. Wei, L. Wang, and A. Y. C. Nee, "Enabling technologies and tools for digital twin," *J. Manuf. Syst.*, vol. 58, pp. 3–21, Jan. 2021. [Online]. Available: <https://api.semanticscholar.org/CorpusID:208841049>
- [19] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital twin in industry: State-of-the-art," *IEEE Trans. Ind. Informat.*, vol. 15, no. 4, pp. 2405–2415, Apr. 2019.
- [20] B. R. Barricelli, E. Casiraghi, and D. Fogli, "A survey on digital twin: Definitions, characteristics, applications, and design implications," *IEEE Access*, vol. 7, pp. 167653–167671, 2019.
- [21] J. R. Bhat and S. A. Alqahtani, "6G ecosystem: Current status and future perspective," *IEEE Access*, vol. 9, pp. 43134–43167, 2021.
- [22] V. K. Quy, A. Chehri, N. M. Quy, N. D. Han, and N. T. Ban, "Innovative trends in the 6G era: A comprehensive survey of architecture, applications, technologies, and challenges," *IEEE Access*, vol. 11, pp. 39824–39844, 2023.
- [23] D. C. Nguyen, M. Ding, P. N. Pathirana, A. Seneviratne, J. Li, D. Niyato, O. Dobre, and H. V. Poor, "6G Internet of Things: A comprehensive survey," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 359–383, Jan. 2022.
- [24] R. Minerva, G. M. Lee, and N. Crespi, "Digital twin in the IoT context: A survey on technical features, scenarios, and architectural models," *Proc. IEEE*, vol. 108, no. 10, pp. 1785–1824, Oct. 2020.
- [25] L. U. Khan, W. Saad, D. Niyato, Z. Han, and C. S. Hong, "Digital-twin-enabled 6G: Vision, architectural trends, and future directions," *IEEE Commun. Mag.*, vol. 60, no. 1, pp. 74–80, Jan. 2022.
- [26] X. Shen, J. Gao, W. Wu, M. Li, C. Zhou, and W. Zhuang, "Holistic network virtualization and pervasive network intelligence for 6G," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 1, pp. 1–30, 1st Quart., 2022.
- [27] F. Tang, X. Chen, T. K. Rodrigues, M. Zhao, and N. Kato, "Survey on digital twin edge networks (DITEN) toward 6G," *IEEE Open J. Commun. Soc.*, vol. 3, pp. 1360–1381, 2022.
- [28] L. U. Khan, Z. Han, W. Saad, E. Hossain, M. Guizani, and C. S. Hong, "Digital twin of wireless systems: Overview, taxonomy, challenges, and opportunities," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 4, pp. 2230–2254, 4th Quart., 2022.
- [29] C. Alcaraz and J. Lopez, "Digital twin: A comprehensive survey of security threats," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 3, pp. 1475–1503, 3rd Quart., 2022.

- [30] B. Han, W. Jiang, M. Asif Habibi, and H. D. Schotten, "An abstracted survey on 6G: Drivers, requirements, efforts, and enablers," 2021, *arXiv:2101.01062*.
- [31] Y. Lu, S. Maharjan, and Y. Zhang, "Adaptive edge association for wireless digital twin networks in 6G," *IEEE Internet Things J.*, vol. 8, no. 22, pp. 16219–16230, Nov. 2021.
- [32] S. Zeb, A. Mahmood, S. A. Hassan, M. J. Piran, M. Gidlund, and M. Guizani, "Industrial digital twins at the Nexus of NextG wireless networks and computational intelligence: A survey," *J. Netw. Comput. Appl.*, vol. 200, Apr. 2021, Art. no. 103309. [Online]. Available: <https://api.semanticscholar.org/CorpusID:236965761>
- [33] M. A. Habibi, B. Han, M. Nasimi, N. P. Kuruvatti, A. Fellan, and H. D. Schotten, "Towards a fully virtualized, cloudified, and slicing-aware RAN for 6G mobile networks," in *6G Mobile Wireless Networks* (Computer Communications and Networks). Cham, Switzerland: Springer, 2021, pp. 327–358.
- [34] M. A. Habibi, F. Z. Yousaf, and H. D. Schotten, "Mapping the VNFs and VLs of a RAN slice onto intelligent PoPs in beyond 5G mobile networks," *IEEE Open J. Commun. Soc.*, vol. 3, pp. 670–704, 2022.
- [35] M. A. Habibi, B. Han, F. Z. Yousaf, and H. D. Schotten, "How should network slice instances be provided to multiple use cases of a single vertical industry?" *IEEE Commun. Standards Mag.*, vol. 4, no. 3, pp. 53–61, Sep. 2020.
- [36] H. Ahmadi, A. Nag, Z. Khar, K. Sayrafian, and S. Rahardja, "Networked twins and twins of networks: An overview on the relationship between digital twins and 6G," *IEEE Commun. Standards Mag.*, vol. 5, no. 4, pp. 154–160, Dec. 2021.
- [37] M. Groshev, C. Guimarães, J. Martín-Pérez, and A. de la Oliva, "Toward intelligent cyber-physical systems: Digital twin meets artificial intelligence," *IEEE Commun. Mag.*, vol. 59, no. 8, pp. 14–20, Aug. 2021.
- [38] P. Agbaje, A. Anjum, A. Mitra, E. Oseghale, G. Bloom, and H. Olufowobi, "Survey of interoperability challenges in the Internet of Vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 22838–22861, Dec. 2022.
- [39] Y. Teng, M. Liu, F. R. Yu, V. C. M. Leung, M. Song, and Y. Zhang, "Resource allocation for ultra-dense networks: A survey, some research issues and challenges," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2134–2168, 3rd Quart., 2019.
- [40] A. Al-Dulaimi, S. Al-Rubaye, J. Cosmas, and A. Anpalagan, "Planning of ultra-dense wireless networks," *IEEE Netw.*, vol. 31, no. 2, pp. 90–96, Mar. 2017.
- [41] A. Laya, K. Wang, A. A. Widaa, J. Alonso-Zarate, J. Markendahl, and L. Alonso, "Device-to-device communications and small cells: Enabling spectrum reuse for dense networks," *IEEE Wireless Commun.*, vol. 21, no. 4, pp. 98–105, Aug. 2014.
- [42] D. Chen and Z. Lv, "Artificial intelligence enabled digital twins for training autonomous cars," *Internet Things Cyber-Physical Syst.*, vol. 2, pp. 31–41, Jan. 2022. [Online]. Available: <https://api.semanticscholar.org/CorpusID:248602008>
- [43] Z. Huang, D. Li, J. Cai, and H. Lu, "Collective reinforcement learning based resource allocation for digital twin service in 6G networks," *J. Netw. Comput. Appl.*, vol. 217, Aug. 2023, Art. no. 103697. [Online]. Available: <https://api.semanticscholar.org/CorpusID:259693938>
- [44] S. Liu, J. Yu, X. Deng, and S. Wan, "FedCPF: An efficient-communication federated learning approach for vehicular edge computing in 6G communication networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 2, pp. 1616–1629, Feb. 2022.
- [45] J. Wang, K. Zhu, and E. Hossain, "Green Internet of Vehicles (IoV) in the 6G era: Toward sustainable vehicular communications and networking," *IEEE Trans. Green Commun. Netw.*, vol. 6, no. 1, pp. 391–423, Mar. 2022.
- [46] K. Liu, X. Xu, P. Dai, and B. Chen, "Cooperative sensing and uploading for quality-cost tradeoff of digital twins in VEC," *IEEE Trans. Consum. Electron.*, 2023, doi: [10.1109/TCE.2023.3318853](https://doi.org/10.1109/TCE.2023.3318853).
- [47] Q. Liu, L. Tang, T. Wu, and Q. Chen, "Deep reinforcement learning for resource demand prediction and virtual function network migration in digital twin network," *IEEE Internet Things J.*, vol. 10, no. 21, pp. 19102–19116, Nov. 2023. [Online]. Available: <https://api.semanticscholar.org/CorpusID>
- [48] M. Sheraz, M. Ahmed, X. Hou, Y. Li, D. Jin, Z. Han, and T. Jiang, "Artificial intelligence for wireless caching: Schemes, performance, and challenges," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 631–661, 1st Quart., 2021.
- [49] M. Sheraz, S. Shafique, S. Imran, M. Asif, R. Ullah, M. Ibrar, J. Khan, and L. Wuttisittikulkij, "A reinforcement learning based data caching in wireless networks," *Appl. Sci.*, vol. 12, no. 11, p. 5692, Jun. 2022. [Online]. Available: <https://api.semanticscholar.org/CorpusID:249352304>
- [50] R. Wang, X. Peng, J. Zhang, and K. B. Letaief, "Mobility-aware caching for content-centric wireless networks: Modeling and methodology," *IEEE Commun. Mag.*, vol. 54, no. 8, pp. 77–83, Aug. 2016.
- [51] C. A. Chan, M. Yan, A. F. Gyax, W. Li, L. Li, I. Chih-Lin, J. Yan, and C. Leckie, "Big data driven predictive caching at the wireless edge," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, May 2019, pp. 1–6.
- [52] Y. Hui, Y. Qiu, N. Cheng, Z. Yin, R. Chen, K. Liang, and T. H. Luan, "Digital twin enabled on-demand content delivery in HetVNs," *IEEE Internet Things J.*, vol. 10, no. 16, pp. 14028–14041, Aug. 2023.
- [53] C. R. Storc and F. Duarte-Figueiredo, "A survey of 5G technology evolution, standards, and infrastructure associated with vehicle-to-everything communications by Internet of Vehicles," *IEEE Access*, vol. 8, pp. 117593–117614, 2020.
- [54] K. Zhang, J. Cao, S. Maharjan, and Y. Zhang, "Digital twin empowered content caching in social-aware vehicular edge networks," *IEEE Trans. Computat. Social Syst.*, vol. 9, no. 1, pp. 239–251, Feb. 2022.
- [55] J. Zheng, T. H. Luan, L. Gao, Y. Zhang, and Y. Wu, "Learning based task offloading in digital twin empowered Internet of Vehicles," 2021, *arXiv:2201.09076*.
- [56] L. Chen, Q. Gu, K. Jiang, and L. Zhao, "A3C-based and dependency-aware computation offloading and service caching in digital twin edge networks," *IEEE Access*, vol. 11, pp. 57564–57573, 2023.
- [57] K. Zhang, J. Cao, and Y. Zhang, "Adaptive digital twin and multi-agent deep reinforcement learning for vehicular edge computing and networks," *IEEE Trans. Ind. Informat.*, vol. 18, no. 2, pp. 1405–1413, Feb. 2022.
- [58] Y. Dai and Y. Zhang, "Adaptive digital twin for vehicular edge computing and networks," *J. Commun. Inf. Netw.*, vol. 7, no. 1, pp. 48–59, Mar. 2022. [Online]. Available: <https://api.semanticscholar.org/CorpusID>
- [59] Y. Lu, X. Huang, K. Zhang, S. Maharjan, and Y. Zhang, "Communication-efficient federated learning and permissioned blockchain for digital twin edge networks," *IEEE Internet Things J.*, vol. 8, no. 4, pp. 2276–2288, Feb. 2021.
- [60] T. Liu, L. Tang, W. Wang, Q. Chen, and X. Zeng, "Digital-twin-assisted task offloading based on edge collaboration in the digital twin edge network," *IEEE Internet Things J.*, vol. 9, no. 2, pp. 1427–1444, Jan. 2022.
- [61] Z. Qu, Y. Li, B. Liu, D. Gupta, and P. Tiwari, "DTQFL: A digital twin-assisted quantum federated learning algorithm for intelligent diagnosis in 5G mobile network," *IEEE J. Biomed. Health Inform.*, doi: [10.1109/JBHI.2023.3303401](https://doi.org/10.1109/JBHI.2023.3303401).
- [62] M. Sheraz, S. Shafique, S. Imran, M. Asif, R. Ullah, M. Ibrar, A. Bartoszewicz, and S. Mobayen, "Mobility-aware data caching to improve D2D communications in heterogeneous networks," *Electronics*, vol. 11, no. 21, p. 3434, Oct. 2022. [Online]. Available: <https://api.semanticscholar.org/CorpusID:253148342>
- [63] M. Ahmed, M. A. Mirza, S. Raza, H. Ahmad, F. Xu, W. U. Khan, Q. Lin, and Z. Han, "Vehicular communication network enabled CAV data offloading: A review," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 8, pp. 7869–7897, Aug. 2023.
- [64] F. Rebecchi, M. Dias de Amorim, V. Conan, A. Passarella, R. Bruno, and M. Conti, "Data offloading techniques in cellular networks: A survey," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 2, pp. 580–603, 2nd Quart., 2015.
- [65] C. Jiang, X. Cheng, H. Gao, X. Zhou, and J. Wan, "Toward computation offloading in edge computing: A survey," *IEEE Access*, vol. 7, pp. 131543–131558, 2019.
- [66] Y. He, J. Guo, and X. Zheng, "From surveillance to digital twin: Challenges and recent advances of signal processing for industrial Internet of Things," *IEEE Signal Process. Mag.*, vol. 35, no. 5, pp. 120–129, Sep. 2018.
- [67] Y. Lu, X. Huang, K. Zhang, S. Maharjan, and Y. Zhang, "Communication-efficient federated learning for digital twin edge networks in industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 17, no. 8, pp. 5709–5718, Aug. 2021.
- [68] W. Sun, H. Zhang, R. Wang, and Y. Zhang, "Reducing offloading latency for digital twin edge networks in 6G," *IEEE Trans. Veh. Technol.*, vol. 69, no. 10, pp. 12240–12251, Oct. 2020.

- [69] Y. Dai, K. Zhang, S. Maharjan, and Y. Zhang, "Deep reinforcement learning for stochastic computation offloading in digital twin networks," *IEEE Trans. Ind. Informat.*, vol. 17, no. 7, pp. 4968–4977, Jul. 2021.
- [70] M. Pengnoo, M. T. Barros, L. Wuttisittikulkij, B. Butler, A. Davy, and S. Balasubramaniam, "Digital twin for metasurface reflector management in 6G terahertz communications," *IEEE Access*, vol. 8, pp. 114580–114596, 2020.
- [71] A. Shafie, N. Yang, and C. Han, "Multi-connectivity for indoor terahertz communication with self and dynamic blockage," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2020, pp. 1–7.
- [72] C.-C. Wang, X.-W. Yao, C. Han, and W.-L. Wang, "Interference and coverage analysis for terahertz band communication in nanonetworks," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2017, pp. 1–6.
- [73] C. Chaccour, M. N. Soorki, W. Saad, M. Bennis, and P. Popovski, "Risk-based optimization of virtual reality over terahertz reconfigurable intelligent surfaces," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2020, pp. 1–6.
- [74] Y. Wang, M. Chen, Z. Yang, W. Saad, T. Luo, S. Cui, and H. V. Poor, "Meta-reinforcement learning for immersive virtual reality over THz/VLC wireless networks," in *Proc. IEEE Int. Conf. Commun.*, Jun. 2021, pp. 1–6.
- [75] Z. Tao, M. Chen, Z. Yang, J. Shi, J. Zhao, and W. Liu, "Optimal control for digital-twin THz/VLC communication networks," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, May 2022, pp. 910–915.
- [76] H. Zhang, X. Ma, X. Liu, L. Li, and K. Sun, "GNN-based power allocation and user association in digital twin network for the terahertz band," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 10, pp. 3111–3121, Oct. 2023.
- [77] C. Huang, S. Hu, G. C. Alexandropoulos, A. Zappone, C. Yuen, R. Zhang, M. D. Renzo, and M. Debbah, "Holographic MIMO surfaces for 6G wireless networks: Opportunities, challenges, and trends," *IEEE Wireless Commun.*, vol. 27, no. 5, pp. 118–125, Oct. 2020.
- [78] B. Sheen, J. Yang, X. Feng, and M. Moin Uddin Chowdhury, "A digital twin for reconfigurable intelligent surface assisted wireless communication," 2020, *arXiv:2009.00454*.
- [79] X. Yuan, J. Chen, N. Zhang, J. Ni, F. R. Yu, and V. C. M. Leung, "Digital twin-driven vehicular task offloading and IRS configuration in the Internet of Vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 24290–24304, Dec. 2022.
- [80] H. R. Hasan, K. Salah, R. Jayaraman, M. Omar, I. Yaqoob, S. Pesic, T. Taylor, and D. Boscovic, "A blockchain-based approach for the creation of digital twins," *IEEE Access*, vol. 8, pp. 34113–34126, 2020.
- [81] Y. Lu, X. Huang, K. Zhang, S. Maharjan, and Y. Zhang, "Low-latency federated learning and blockchain for edge association in digital twin empowered 6G networks," *IEEE Trans. Ind. Informat.*, vol. 17, no. 7, pp. 5098–5107, Jul. 2021.
- [82] C. Altun, B. Tavli, and H. Yanikomeroglu, "Liberalization of digital twins of IoT-enabled home appliances via blockchains and absolute ownership rights," *IEEE Commun. Mag.*, vol. 57, no. 12, pp. 65–71, Dec. 2019.
- [83] B. Li, Y. Liu, L. Tan, H. Pan, and Y. Zhang, "Digital twin assisted task offloading for aerial edge computing and networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 10, pp. 10863–10877, Oct. 2022.
- [84] W. Liu, B. Li, W. Xie, Y. Dai, and Z. Fei, "Energy efficient computation offloading in aerial edge networks with multi-agent cooperation," *IEEE Trans. Wireless Commun.*, vol. 22, no. 9, pp. 5725–5739, Sep. 2023.
- [85] X. Tang, X. Li, R. Yu, Y. Wu, J. Ye, and F. Tang, "Digital twin assisted task assignment in multi-UAV systems: A deep reinforcement learning approach," *IEEE Internet Things J.*, vol. 10, no. 17, pp. 15362–15375, Sep. 2023.
- [86] A. Soliman, A. Al-Ali, A. Mohamed, H. Gedawy, D. Izham, M. Bahri, A. Erbad, and M. Guizani, "AI-based UAV navigation framework with digital twin technology for mobile target visitation," *Eng. Appl. Artif. Intell.*, vol. 123, Aug. 2023, Art. no. 106318. [Online]. Available: <https://api.semanticscholar.org/CorpusID:258434958>
- [87] Y. Qu, L. Gao, Y. Xiang, S. Shen, and S. Yu, "FedTwin: Blockchain-enabled adaptive asynchronous federated learning for digital twin networks," *IEEE Netw.*, vol. 36, no. 6, pp. 183–190, Nov. 2022.
- [88] M. Mozaffari, W. Saad, M. Bennis, Y.-H. Nam, and M. Debbah, "A tutorial on UAVs for wireless networks: Applications, challenges, and open problems," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2334–2360, 3rd Quart., 2019.
- [89] F. Dembski, U. Wössner, M. Letzgus, M. Ruddat, and C. Yamu, "Urban digital twins for smart cities and citizens: The case study of Herrenberg, Germany," *Sustainability*, vol. 12, no. 6, p. 2307, Mar. 2020. [Online]. Available: <https://www.mdpi.com/2071-1050/12/6/2307>
- [90] F. Dembski, C. Yamu, and U. Wössner, "Digital twin, virtual reality and space syntax: Civic engagement and decision support for smart, sustainable cities," in *Proc. 12th Int. Space Syntax Symp.*, 2019, pp. 1–14. [Online]. Available: <https://api.semanticscholar.org/CorpusID:197725347>
- [91] A. Deckert, F. Dembski, F. Ulmer, M. Ruddat, and U. Wössner, "Digital tools in stakeholder participation for the German energy transition. Can digital tools improve participation and its outcome?" in *The Role of Public Participation in Energy Transitions*. Cambridge, MA, USA: Academic, 2020.
- [92] T. Deng, K. Zhang, and Z. J. M. Shen, "A systematic review of a digital twin city: A new pattern of urban governance toward smart cities," *J. Manag. Sci. Eng.*, vol. 6, no. 2, pp. 125–134, 2021. [Online]. Available: <https://api.semanticscholar.org/CorpusID:233678798>
- [93] S. Shirowzhan, W. Tan, and S. M. E. Sepasgozar, "Digital twin and CyberGIS for improving connectivity and measuring the impact of infrastructure construction planning in smart cities," *ISPRS Int. J. Geo-Inf.*, vol. 9, no. 4, p. 240, Apr. 2020. [Online]. Available: <https://api.semanticscholar.org/CorpusID:216266776>
- [94] F. Dembski, U. Wössner, M. Letzgus, M. Ruddat, and C. Yamu, "Urban digital twins for smart cities and citizens: The case study of herrenberg, Germany," *Sustainability*, vol. 12, no. 6, p. 2307, Mar. 2020. [Online]. Available: <https://api.semanticscholar.org/CorpusID:216404355>
- [95] A. A. Salunke, "Reinforcement learning empowered digital twins: Pioneering smart cities towards optimal urban dynamics," *EPRA Int. J. Res. Develop. (IJRD)*, vol. 8, no. 7, pp. 265–269, 2023. [Online]. Available: <https://api.semanticscholar.org/CorpusID:260393398>
- [96] P. Bellini, C. M. Colombo, M. Fanfani, P. Nesi, G. Pantaleo, R. Pisanu, and L. Adreani, "Digital twin framework for smart city solutions," in *Proc. Int. Distrib. Multimedia Syst. Conf. Vis. Vis. Lang.*, 2022, pp. 1–8. [Online]. Available: <https://api.semanticscholar.org/CorpusID:251776924>
- [97] Z. Pervez, Z. Khan, A. Ghafoor, and K. Soomro, "SIGNED: Smart city digital twin verifiable data framework," *IEEE Access*, vol. 11, pp. 29430–29446, 2023.
- [98] M. Zhou, J. Yan, and D. Feng, "Digital twin framework and its application to power grid online analysis," *CSEE J. Power Energy Syst.*, vol. 5, no. 3, pp. 391–398, Sep. 2019. [Online]. Available: <https://api.semanticscholar.org/CorpusID>
- [99] T. R. Wanasinghe, L. Wroblewski, B. K. Petersen, R. G. Gosine, L. Anne James, O. De Silva, G. K. I. Mann, and P. J. Warrian, "Digital twin for the oil and gas industry: Overview, research trends, opportunities, and challenges," *IEEE Access*, vol. 8, pp. 104175–104197, 2020.
- [100] M. Zhang, F. Tao, and A. Y. C. Nee, "Digital twin enhanced dynamic job-shop scheduling," *J. Manuf. Syst.*, vol. 58, pp. 146–156, Jan. 2021. [Online]. Available: <https://api.semanticscholar.org/CorpusID:219051787>
- [101] Q. Lu, X. Xie, A. K. Parlikad, and J. M. Schooling, "Digital twin-enabled anomaly detection for built asset monitoring in operation and maintenance," *Autom. Construct.*, vol. 118, Oct. 2020, Art. no. 103277. [Online]. Available: <https://api.semanticscholar.org/CorpusID:219750657>
- [102] Q. Qiao, J. Wang, L. Ye, and R. X. Gao, "Digital twin for machining tool condition prediction," *Proc. CIRP*, vol. 81, pp. 1388–1393, Jan. 2019. [Online]. Available: <https://api.semanticscholar.org/CorpusID:198340366>
- [103] D. Guivarch, E. Mermoz, Y. Marino, and M. Sartor, "Creation of helicopter dynamic systems digital twin using multibody simulations," *CIRP Ann.*, vol. 68, no. 1, pp. 133–136, 2019. [Online]. Available: <https://api.semanticscholar.org/CorpusID:155356444>
- [104] L. Pan, X. Guo, Y. Luan, and H. Wang, "Design and realization of cutting simulation function of digital twin system of CNC machine tool," *Proc. Comput. Sci.*, vol. 183, pp. 261–266, Jan. 2021. [Online]. Available: <https://api.semanticscholar.org/CorpusID:234964978>
- [105] A. A. Malik and A. Bilberg, "Digital twins of human robot collaboration in a production setting," *Proc. Manuf.*, vol. 17, pp. 278–285, Jan. 2018. [Online]. Available: <https://api.semanticscholar.org/CorpusID:69375263>

- [106] A. Khalid, P. Kirisci, Z. H. Khan, Z. Ghrairi, K.-D. Thoben, and J. Pannek, "Security framework for industrial collaborative robotic cyber-physical systems," *Comput. Ind.*, vol. 97, pp. 132–145, May 2018. [Online]. Available: <https://api.semanticscholar.org/CorpusID:46918629>
- [107] M. Chen, J. Shao, S. Guo, L. Su, and H. Du, "Convoy_DTN: A security interaction engine design for digital twin network," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, Dec. 2021, pp. 1–5.
- [108] L. Guo, Y. Cheng, Y. Zhang, Y. Liu, C. Wan, and J. Liang, "Development of cloud-edge collaborative digital twin system for FDM additive manufacturing," in *Proc. IEEE 19th Int. Conf. Ind. Informat. (INDIN)*, Jul. 2021, pp. 1–6.
- [109] F. Laamarti, H. F. Badawi, Y. Ding, F. Arafsha, B. Hafidh, and A. E. Saddik, "An ISO/IEEE 11073 standardized digital twin framework for health and well-being in smart cities," *IEEE Access*, vol. 8, pp. 105950–105961, 2020.
- [110] F. Firouzi, B. Farahani, M. Daneshmand, K. Grise, J. Song, R. Saracco, and L. L. Wang, "Harnessing the power of smart and connected health to tackle COVID-19: IoT, AI, robotics, and blockchain for a better world," *IEEE Internet Things J.*, vol. 8, no. 16, pp. 12826–12846, Aug. 2021.
- [111] H. Elayan, M. Aloqaily, and M. Guizani, "Digital twin for intelligent context-aware IoT healthcare systems," *IEEE Internet Things J.*, vol. 8, no. 23, pp. 16749–16757, Dec. 2021.
- [112] M. B. Mollah, J. Zhao, D. Niyato, Y. L. Guan, C. Yuen, S. Sun, K.-Y. Lam, and L. H. Koh, "Blockchain for the Internet of Vehicles towards intelligent transportation systems: A survey," *IEEE Internet Things J.*, vol. 8, no. 6, pp. 4157–4185, Mar. 2021.
- [113] T. K. Rodrigues, K. Suto, H. Nishiyama, J. Liu, and N. Kato, "Machine learning meets computation and communication control in evolving edge and cloud: Challenges and future perspective," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 1, pp. 38–67, 1st Quart., 2020.
- [114] R. Saracco, "Digital twins: Bridging physical space and cyberspace," *Computer*, vol. 52, no. 12, pp. 58–64, Dec. 2019. [Online]. Available: <https://api.semanticscholar.org/CorpusID>
- [115] X. Liao, Z. Wang, X. Zhao, K. Han, P. Tiwari, M. J. Barth, and G. Wu, "Cooperative ramp merging design and field implementation: A digital twin approach based on vehicle-to-cloud communication," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 5, pp. 4490–4500, May 2022.
- [116] Z. Wang, K. Han, and P. Tiwari, "Digital twin-assisted cooperative driving at non-signalized intersections," *IEEE Trans. Intell. Vehicles*, vol. 7, no. 2, pp. 198–209, Jun. 2022.
- [117] X. Chen, E. Kang, S. Shiraishi, V. M. Preciado, and Z. Jiang, "Digital behavioral twins for safe connected cars," in *Proc. 21st ACM/IEEE Int. Conf. Model Driven Eng. Lang. Syst.*, Oct. 2018, pp. 144–153. [Online]. Available: <https://api.semanticscholar.org/CorpusID:52936668>
- [118] S. A. P. Kumar, R. Madhumathi, P. R. Chelliah, L. Tao, and S. Wang, "A novel digital twin-centric approach for driver intention prediction and traffic congestion avoidance," *J. Reliable Intell. Environments*, vol. 4, no. 4, pp. 199–209, Dec. 2018. [Online]. Available: <https://api.semanticscholar.org/CorpusID:53085686>
- [119] M. González, O. Salgado, J. Croes, B. Plummers, and W. Desmet, "A digital twin for operational evaluation of vertical transportation systems," *IEEE Access*, vol. 8, pp. 114389–114400, 2020.
- [120] S. Liao, J. Wu, A. K. Bashir, W. Yang, J. Li, and U. Tariq, "Digital twin consensus for blockchain-enabled intelligent transportation systems in smart cities," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22619–22629, Nov. 2022.
- [121] X. Liao, X. Zhao, Z. Wang, Z. Zhao, K. Han, R. Gupta, M. J. Barth, and G. Wu, "Driver digital twin for online prediction of personalized lane change behavior," *IEEE Internet Things J.*, vol. 10, no. 15, pp. 13235–13246, Aug. 2023.
- [122] S. Dasgupta, M. Rahman, A. D. Lidbe, W. Lu, and S. Jones, "A transportation digital-twin approach for adaptive traffic control systems," 2021, *arXiv:2109.10863*.
- [123] N. Bazmohammadi, A. Madary, J. C. Vasquez, H. B. Mohammadi, B. Khan, Y. Wu, and J. M. Guerrero, "Microgrid digital twins: Concepts, applications, and future trends," *IEEE Access*, vol. 10, pp. 2284–2302, 2022.
- [124] S. Suhail, R. Hussain, R. Jurdak, A. Oracevic, K. Salah, C. S. Hong, and R. Matulevičius, "Blockchain-based digital twins: Research trends, issues, and future challenges," *ACM Comput. Surv.*, vol. 54, no. 11s, pp. 1–34, 2022.
- [125] Q. Guo, F. Tang, and N. Kato, "Federated reinforcement learning-based resource allocation for D2D-aided digital twin edge networks in 6G industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 19, no. 5, pp. 7228–7236, May 2023.
- [126] H. Xu, J. Wu, J. Li, and X. Lin, "Deep-reinforcement-learning-based cyber-twin architecture for 6G IIoT: An integrated design of control, communication, and computing," *IEEE Internet Things J.*, vol. 8, no. 22, pp. 16337–16348, Nov. 2021.
- [127] W. Yang, W. Xiang, Y. Yang, and P. Cheng, "Optimizing federated learning with deep reinforcement learning for digital twin empowered industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 19, no. 2, pp. 1884–1893, Feb. 2023.
- [128] A. Islam and S. Y. Shin, "A digital twin-based drone-assisted secure data aggregation scheme with federated learning in artificial intelligence of things," *IEEE Netw.*, vol. 37, no. 2, pp. 278–285, Mar./Apr. 2023.
- [129] K. B. Letaief, Y. Shi, J. Lu, and J. Lu, "Edge artificial intelligence for 6G: Vision, enabling technologies, and applications," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 1, pp. 5–36, Jan. 2022.
- [130] A. Rasheed, O. San, and T. Kvamsdal, "Digital twin: Values, challenges and enablers from a modeling perspective," *IEEE Access*, vol. 8, pp. 21980–22012, 2020.
- [131] C. Zhu, X. Zhu, J. Ren, and T. Qin, "Blockchain-enabled federated learning for UAV edge computing network: Issues and solutions," *IEEE Access*, vol. 10, pp. 56591–56610, 2022.
- [132] G. Zhang, V. Davoodnia, A. Sepas-Moghaddam, Y. Zhang, and A. Etemad, "Classification of hand movements from EEG using a deep attention-based LSTM network," *IEEE Sensors J.*, vol. 20, no. 6, pp. 3113–3122, Mar. 2020.
- [133] Z. Mahrez, E. Sabir, E. Badidi, W. Saad, and M. Sadik, "Smart urban mobility: When mobility systems meet smart data," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6222–6239, Jul. 2022.
- [134] Y. He, M. Chen, B. Ge, and M. Guizani, "On WiFi offloading in heterogeneous networks: Various incentives and trade-off strategies," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 4, pp. 2345–2385, 4th Quart., 2016.
- [135] Z. Lv, C. Cheng, and H. Lv, "Blockchain-based decentralized learning for security in digital twins," *IEEE Internet Things J.*, vol. 10, no. 24, pp. 21479–21488, 2023.
- [136] J. Lee, J. Kim, I. Kim, and K. Han, "Cyber threat detection based on artificial neural networks using event profiles," *IEEE Access*, vol. 7, pp. 165607–165626, 2019.
- [137] Y. Wang, Z. Su, S. Guo, M. Dai, T. H. Luan, and Y. Liu, "A survey on digital twins: Architecture, enabling technologies, security and privacy, and future prospects," *IEEE Internet Things J.*, vol. 10, no. 17, pp. 14965–14987, Sep. 2023.
- [138] K. Azbeg, O. Ouchetto, and S. J. Andaloussi, "Access control and privacy-preserving blockchain-based system for diseases management," *IEEE Trans. Computat. Social Syst.*, vol. 10, no. 4, pp. 1515–1527, Aug. 2023.
- [139] R. Vieira, L. Gonçalves, and F. Schiavoni, "The things of the internet of musical things: Defining the difficulties to standardize the behavior of these devices," in *Proc. X Brazilian Symp. Comput. Syst. Eng. (SBESC)*, Nov. 2020, pp. 1–7.
- [140] C. Tan, X. Li, L. Gao, T. H. Luan, Y. Qu, Y. Xiang, and R. Lu, "Digital twin enabled remote data sharing for Internet of Vehicles: System and incentive design," *IEEE Trans. Veh. Technol.*, vol. 72, no. 10, pp. 13474–13489, Oct. 2023.
- [141] M. Pradhan, N. Suri, C. Fuchs, T. H. Bloebaum, and M. Marks, "Toward an architecture and data model to enable interoperability between federated mission networks and IoT-enabled smart city environments," *IEEE Commun. Mag.*, vol. 56, no. 10, pp. 163–169, Oct. 2018.
- [142] M. Mongelli and S. Scanzio, "Approximating optimal estimation of time offset synchronization with temperature variations," *IEEE Trans. Instrum. Meas.*, vol. 63, no. 12, pp. 2872–2881, Dec. 2014.
- [143] D. T. Green, "Hipa security rule compliance in small healthcare facilities: A theoretical framework," *Issues Inf. Syst.*, vol. 16, no. 1, pp. 180–188, 2015. [Online]. Available: <https://api.semanticscholar.org/CorpusID:218465213>
- [144] R. Morabito, V. Cozzolino, A. Y. Ding, N. Bejjar, and J. Ott, "Consolidate IoT edge computing with lightweight virtualization," *IEEE Netw.*, vol. 32, no. 1, pp. 102–111, Jan. 2018.

- [145] Q. Chen, Z. Zheng, C. Hu, D. Wang, and F. Liu, "On-edge multi-task transfer learning: Model and practice with data-driven task allocation," *IEEE Trans. Parallel Distrib. Syst.*, vol. 31, no. 6, pp. 1357–1371, Jun. 2020.
- [146] L. U. Khan, E. Mustafa, J. Shuja, F. Rehman, K. Bilal, Z. Han, and C. S. Hong, "Federated learning for digital twin-based vehicular networks: Architecture and challenges," *IEEE Wireless Commun.*, early access, Mar. 7, 2023, doi: [10.1109/MWC.012.2200373](https://doi.org/10.1109/MWC.012.2200373).
- [147] M. Chen, Z. Yang, W. Saad, C. Yin, H. V. Poor, and S. Cui, "A joint learning and communications framework for federated learning over wireless networks," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 269–283, Jan. 2021.



intelligent transportation systems (ITS), and Industry 4.0. He has served as a TPC member for several conferences.

MUHAMMAD SHERAZ is a Research Fellow with the Faculty of Engineering, Multimedia University, Malaysia. His research interests include information-theoretic aspects of wireless networks, signal processing, mmWave and THz communications, AI, digital twin networks, resource allocation, non-orthogonal multiple access (NOMA), reconfigurable intelligent surfaces (RIS), ultra-reliable low latency communications (URLLC), edge computing, the IoTs, intelligent transportation systems (ITS), and Industry 4.0. He has served as a TPC member for several conferences.



TEONG CHEE CHUAH received the B.Eng. and Ph.D. degrees from Newcastle University, U.K., in 1999 and 2002, respectively. He is currently a Professor with the Faculty of Engineering, Multimedia University, Malaysia. His research interests include resource management for fixed-mobile convergent networks, applications of artificial intelligence in communications, and cellular network planning and optimization.



YING LOONG LEE (Senior Member, IEEE) received the B.Eng. (Hons.) degree in electronics majoring in telecommunications and the Ph.D. degree in engineering from Multimedia University, Malaysia, in 2012 and 2017, respectively. He was a Visiting Researcher with South China University of Technology, China, in 2023; the Technical University of Darmstadt, Germany, in 2022; Guangxi University for Nationalities, China; and National Chiao Tung University, Taiwan, in 2015. He is currently an Assistant Professor with the Department of Electrical and Electronic Engineering, Lee Kong Chian Faculty of Engineering and Science, Universiti Tunku Abdul Rahman. His current research interests include wireless communications for 6G and beyond, fixed-mobile convergence, and applications of artificial intelligence in telecommunications. He was a recipient of the IEEE Malaysia ComSoc/VTS Joint Chapter Outstanding Doctoral Dissertation Award, in 2018, and the Multimedia University Best Thesis Award for Ph.D. (Engineering), in 2017. He was also a co-recipient of the IEEE Vehicular Technology Society OJVT Best Paper Award, in 2023, for his journal article entitled "6G Massive Radio Access Networks: Key Applications, Requirements and Challenges" published in IEEE OPEN JOURNAL OF VEHICULAR TECHNOLOGY.



MUHAMMAD MAHTAB ALAM (Senior Member, IEEE) received the M.Sc. degree in electrical engineering from Aalborg University, Denmark, in 2007, and the Ph.D. degree in signal processing and telecommunication from the INRIA Research Center, University of Rennes I, France, in 2013. He joined the Swedish College of Engineering and Technology, Pakistan, as an Assistant Professor, in 2013. From 2014 to 2016, he was a Postdoctoral Researcher with Qatar Mobility Innovation Center, Qatar. In 2016, he joined as the European Research Area (ERA) Chair and an Associate Professor with the Thomas Johann Seebeck Department of Electronics, Tallinn University of Technology, where he was elected as a Professor, in 2018. In 2019, he became the Head of the Communication Systems Research Group Leader and led a number of national and international projects. His research interests include wireless communications and connectivity, NB-IoT 5G/B5G smart networks and services, and low-power wearable networks for SmartHealth.



ALA'A AL-HABASHNA received the M.Eng. degree from the Memorial University of Newfoundland, in 2010, and the Ph.D. degree in electrical and computer engineering from Carleton University, Ottawa, ON, Canada, in 2018. He is currently an Assistant Professor with Al Hussein Technical University, Amman, Jordan, and an Adjunct Research Professor with Carleton University. His current research interests include 5G wireless networks, machine learning, computer vision, the IoT applications, localization, multimedia communication over wireless networks, signal detection and classification, cognitive radio systems, and discrete-event modeling and simulation. He has won multiple awards, including excellence and best paper awards. He was a reviewer for many conferences and journals and a technical program committee member of multiple conferences. He has been the Co-Chair of the Communications and Networking Simulation (CNS) Track of the Annual Modeling and Simulation Conference.



ZHU HAN (Fellow, IEEE) received the B.S. degree in electronic engineering from Tsinghua University in 1997 and the M.S. and Ph.D. degrees in electrical and computer engineering from the University of Maryland, College Park, MD, USA, in 1999 and 2003, respectively. From 2000 to 2002, he was a Research and Development Engineer with JDSU, Germantown, Maryland. From 2003 to 2006, he was a Research Associate with the University of Maryland. From 2006 to 2008, he was an Assistant Professor with Boise State University, Boise, ID, USA. He is currently a Professor with the Department of Electrical and Computer Engineering and the Department of Computer Science, University of Houston, Houston, TX, USA. His research interests include wireless resource allocation and management, wireless communications and networking, game theory, big data analysis, security, and smart grids. He received an NSF Career Award, in 2010; the Fred W. Ellersick Prize of the IEEE Communication Society, in 2011; the Best Paper Award for the *EURASIP Journal on Advances in Signal Processing*, in 2015; IEEE Leonard G. Abraham Prize in the field of communications systems (Best Paper Award in IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS) in 2016; and several best paper awards in IEEE conferences. He is an IEEE Communications Society Distinguished Lecturer.

...